

Who bears the trade shocks?

Distributional incidence and fiscal cushioning in the UK*

Vahid Ahmadi[†]

August 2026

Abstract

This paper simulates two price-led shocks, the post-Brexit food price rise and the 2022 energy surge, through the UK tax and benefit system in PolicyEngine UK, within a dossier of five trade events of 2021–26. Event magnitudes are imported from published estimates; results are conditional statutory incidence, not causal effects. Both shocks fell hardest on lower-income deciles as a share of spending. With nominal incomes fixed, the fourteen modelled cash benefits did not change; the model’s Energy Price Guarantee variable responded by £1.8 billion, classified as discretionary. A rulebook-vintage run puts the crisis-year cost of lagged uprating at roughly £15 billion, alongside other parameter changes between rule years. Modelled energy-specific payments equalled 49.4 to 63.8 per cent of the counterfactual shock (60.2 to 79.8 including cost-of-living payments); no instrument was legislated against the food price rise. On imputed exposure, 93.3 per cent of energy-burden variance lies within income deciles, inheriting the imputation’s dispersion.

Keywords: trade shocks, tariffs, Brexit, terms of trade, microsimulation, automatic stabilisers, public finances

JEL codes: F13, F14, F16, D31, H23, I38

***Acknowledgements:** I thank Max Ghenis and Seyed Mahdi Hosseini for valuable comments and helpful discussions.

[†]Research Associate at PolicyEngine. Email: vahid@policyengine.org.

1 Introduction

Britain has just lived through an unusually dense sequence of trade events: three acts of trade policy (a new customs border with its largest trading partner, 2021; a tariff war with its largest single-country export market, 2025; and a new trade agreement with India, 2026); the sharpest trade-transmitted event since the 1970s, the 2022 terms-of-trade deterioration; and one near-zero agreement (CPTPP, 2024) that serves as a benchmark. The policy/market distinction is carried explicitly throughout: the energy episode is not a policy act, and the comparison never treats it as one. Each episode has its own literature, and several have been studied at household level: the TCA’s food-price effect by income decile, the energy shock’s incidence and relief in detail. What is missing is the join: none has been put through the statutory tax–benefit system that determines what households finally keep, and none has been compared with the others under a common lens. This paper supplies the join in full for the two largest price episodes, and the comparative lens, with its conditions declared, for all five.

The design holds the second stage fixed and varies the shock. Each episode enters as a *declared first stage* (published estimates for the negative shocks, the government’s own projections for the positive ones, never re-estimated here) and is mapped onto households through two channels: consumer prices against expenditure patterns, and labour income through the statutory tax–benefit system. Because gross sizes span two orders of magnitude, comparisons run in ratios: burden as a share of spending by income decile and cushioning shares. And because the rulebook itself moved across the window (the Covid uplift removed in 2021; upratings lagging a year behind prices; the Energy Price Guarantee legislated mid-shock), the design treats the tax–benefit vintage as an explicit dimension: the same shock is scored under the rules of its day, under the previous year’s rules, and with the discretionary layer switched off.

The first-pass results already qualify the received wisdom in three ways. First, the negative shocks were regressive and the positive episodes were not progressive: on its realised path the energy shock took 5.93 per cent of bottom-decile spending against 2.76 per cent at the top. The TCA food-price effect was 0.35 against 0.20 per cent on the window-average path, while the India agreement’s consumer gains are proportional and worth a few pounds a year, too small to register as a share of spending. Second, each episode’s regressivity is generated by its category’s budget-share gradient: food takes 14.9 per cent of bottom-decile spending against 8.3 at the top, energy 7.0 against 3.2, clothing and footwear roughly equal shares everywhere: composition, not size, decides who bears a trade shock. Third, the finding the rulebook axis exists for: the modelled responses run through different instruments on different margins. The earnings-led tariff scenario was cushioned automatically (36.6 to 43.9 per cent, from the companion study); the price-led energy shock was cushioned by discretion, the Guarantee alone taking 33.8 per cent of the counterfactual shock; and indexation, the price side’s one automatic channel, followed the previous September’s inflation, leaving a single adult’s Universal Credit allowance £393 below contemporaneous indexation in the crisis year, months after the £1,040 uplift ended. The two margins carry different estimands, as the anatomy table’s notes make explicit.

The contribution is threefold: a statutory study of Britain’s two major price-led trade shocks, embedded in a five-episode comparative dossier that states the conditions every episode’s numbers carry. First, the statutory second stage itself. The trade literature measures shocks without the system that redistributes them; the microsimulation literature measures the system against income shocks, for which it is designed. Joining them for the energy and post-Brexit food episodes shows that, with nominal incomes fixed, the modelled cash benefits do not respond to a *price* shock (the model’s own Energy Price Guarantee variable does, and the paper counts it

as discretionary), that the discretionary response to one shock was large and cushioned lower deciles at higher rates while the other received no instrument, and that, on imputed exposure, most of the variance in the energy burden lies within income deciles rather than between them (Section 5); a closing exercise aggregates the measured incidence into MPC-weighted first-round demand and Atkinson-weighted welfare under declared parameters (Section 5.8). Second, the rulebook axis: holding a shock fixed and varying the vintage of the welfare state that receives it. It is executed here for the largest episode, the same households scored under the 2022 and 2023 parameter systems, which prices the crisis-year uprating lag through the full statutory structure (Section 4.3), and specified for the rest (Section 6.1); post-Covid Britain, four non-benchmark episodes landing on four rulebooks, is its natural laboratory. Third, the comparative dossier, offered as context rather than as the delivered analysis: the three remaining episodes are carried at decile level or as declared benchmarks, every first stage is labelled estimate or projection, every dial is declared, and one episode (CPTPP) is carried as a near-zero benchmark, reported only as an aggregate divided by a household count, with no distribution assigned. The dossier cannot rank episodes on welfare and its cross-episode ratios are descriptive; it records that, on an episode-specific attribution, the energy shock was met with energy-specific instruments and the food price rise with none.

The paper proceeds in reading order. Section 3 defines the estimand, states what does and does not require identification, sets out the five episodes with their declared first stages, and describes the public-data first pass. Section 4 reports the first pass; Section 5 runs the two largest episodes, the energy shock and the TCA food price rise, through the statutory system on enhanced-FRS household records in PolicyEngine UK, where the automatic-versus-discretionary decomposition, the rulebook-vintage run, the within-decile dispersion and the demand and welfare aggregation are computed. Section 6 concludes and specifies the extension to the remaining episodes (Section 6.1).

2 Literature

Four literatures meet here without previously having met each other: the household incidence of trade policy, the microsimulation of shocks through statutory tax–benefit systems, the distributional analysis of price shocks and benefit indexation, and the UK’s episode-specific empirical record since 2020. Each has done part of what this paper does; none has joined them.

2.1 Trade to households

The framework for pushing trade shocks onto household microdata is best developed for developing countries: the World Bank’s Household Impacts of Tariffs programme simulates tariff changes through both expenditure and income portfolios for 54 countries (Artuç et al., 2021, 2019), and the OECD has published the concordance machinery for mapping CGE trade-model output onto household-budget-survey categories (Luu et al., 2020). The structural anchor for the consumption side is Fajgelbaum and Khandelwal (2016): trade’s gains flow disproportionately to lower-income households through traded-goods budget shares. The 2025 tariff war produced a fast-moving US applied literature, the Budget Lab’s rolling distributional scoring of the enacted schedules and the Tax Policy Center’s implementation of tariffs inside a working tax microsimulation model (Budget Lab, 2025; TPC, 2025), and an academic treatment of two-sided incidence on household expenditure microdata (Fleck and Pradhan, 2026). The CGE–microsimulation linkage tradition supplies the architecture for putting a trade model upstairs and microdata

downstairs (Robilliard and Robinson, 2005; Peichl, 2008; Colombo, 2010; Cali et al., 2019); of that tradition, only Peichl (2008) carries actual statutory tax rules in the micro layer.

What none of this work contains is a tax-benefit system responding to the shock. The World Bank and OECD frameworks stop at market incomes and expenditure; the US 2025 work is consumption-side; the CGE-linkage papers (Peichl’s flat-tax application aside) use reduced-form income equations. Among the studies reviewed here, no shock arriving under a *trade* label has been run through a statutory tax-benefit calculator, EUROMOD, UKMOD, TAXBEN, PolicyEngine or any counterpart, other than the companion study discussed below and Section 5 of this paper, which carries both the energy episode and the TCA food-price episode, a trade-policy act, through the statutory stage. The claim is deliberately narrow: price and energy shocks as such have been run through these models (the close antecedents are discussed in the next subsection), and what is new is the trade first stage and the episode-comparative design, not the machinery.

2.2 Shocks through statutory rules

That machinery exists and has been applied to every kind of shock except trade. Dolls et al. (2012) measure the stabilisation coefficients of the US and European systems against stylised income and unemployment shocks; Brewer and Tasseva (2021) impose the observed Covid labour-market shock on FRS microdata through UKMOD and decompose the cushioning into automatic and discretionary parts; the EUROMOD nowcasting tradition supplies the labour-transition machinery (Navicke et al., 2014). Two studies come close enough to this paper’s rulebook axis that the claim must be stated narrowly. Amores et al. (2025) run the 2022 inflation shock and its offsetting fiscal measures through EUROMOD for the EU, separating income-support from price-containing instruments (in effect a price-shock stabilisation coefficient, and the direct antecedent of what this paper computes); they find fiscal measures recovered roughly a third of welfare losses while targeting them poorly. Popova et al. (2026) are closer still on the UK: UKMOD with LCFS consumption imputed onto the FRS, disposable and consumable income, and an explicit CPI-versus-earnings indexation counterfactual over 2019–2023. Neither models the September-CPI *timing* lag, the Energy Price Guarantee, or any trade shock, and neither treats automatic stabilisation in anything but the classic income-shock sense. That quartet, timing lag, discretionary energy layer, trade first stages, and the comparison across episodes, is what this paper adds, and it is a narrower claim than “nobody has measured price-shock stabilisation.”

The conceptual gap it rests on is old and explicit: simulating an oil-price supply shock in FRB/US, Cohen and Follette (2000) found that automatic stabilisers do very little against a supply shock, unlike their demand-shock performance, a result reported in a macro model rather than on microdata. Paulus et al. (2020) supply the canonical treatment of non-indexation as a distributional phenomenon, and Leventi et al. (2024) the nearest existing treatment of indexation itself as a stabiliser. The companion study (Ahmadi, 2026), an unpublished manuscript, applies tax-benefit microsimulation to a trade shock, covering a single episode (the 2025 US tariffs) through the earnings channel only. This paper deepens it on the price margin: two priced episodes through the statutory stage, a five-episode comparative dossier, and the rulebook vintage as an explicit dimension.

2.3 Price shocks, indexation and the rulebook

The paper’s rulebook axis has more precedent than its trade axis, and the debts should be stated plainly. Cribb et al. (2023) quantified the 2022–23 uprating shortfall directly (benefits rose 3.1 per cent against roughly ten per cent inflation; recipients were not made whole until April 2025) and scored the flat cost-of-living payments against a timely-indexation counterfactual, finding the discretionary route cost more while leaving hundreds of thousands of means-tested households worse off than indexation would have. OBR (2022b) puts the aggregate real fall at around five per cent, some £12 billion, in 2022–23. Judge and Murphy (2023) supply the fact that pre-empt the obvious objection that lags wash out over a cycle: across fifteen years, working-age benefits were uprated below outturn inflation in ten fiscal years, so the asymmetry is empirical, not assumed. Internationally, Eckerstorfer et al. (2025) microsimulate Austria’s inflation episode with lagged wage and pension adjustment, the closest non-UK analogue. This paper adds neither the discovery of the lag nor a better estimate of it; what it adds is the join, the same lag priced against *trade-shock* incidence, so that the timing of indexation and the import content of a household’s basket are evaluated in one place, across episodes rather than one.

One contrary claim must be met rather than cited in passing. Jaravel (2021) concludes that trade affects price levels roughly similarly across income groups, innovation rather than trade driving inflation inequality. The episodes here are consistent with that as a statement about trade *in general* and inconsistent with it as a statement about these shocks in particular: what makes the UK’s negative episodes regressive is not trade exposure as such but the categories they landed on, food and domestic energy, whose budget shares fall steeply with income (Figure 3). Where a trade shock lands on clothing, as the India agreement does, this paper finds exactly the flat incidence Jaravel describes.

The method template for the consumption channel predates all of this: Cravino and Levchenko (2017) showed after Mexico’s 1994 devaluation that lower-income households’ cost of living rose about half again as much as higher-income households’, because they spend more on tradeables and buy cheaper varieties within categories, the second mechanism being one that category-level data cannot see. The decomposition machinery for price shocks is also established: Sologon et al. (2025) decompose inflation’s welfare and distributional impact across expenditure patterns, and Chen et al. (2024) map an exchange-rate shock onto heterogeneous household price experiences using scanner data. Their finding is a direct bound on this paper’s first pass: inflation inequality surged in 2021–23 partly because cheaper goods inflated fastest, a within-category quality margin that published expenditure-by-decile tables hold fixed and therefore understate. On the policy question the episodes raise, the sharpest numbers are Italian: Curci et al. (2025) find targeted social bonuses costing a fifth as much as untargeted VAT and excise cuts drove most of the inequality reduction, with nearly half of untargeted energy support accruing above the median.

On the price-response side, Levell et al. (2025) is the frontier and a standing challenge to any decile-based exercise: using transaction data they find income explains only a small share of the variation in energy spending, so within-decile dispersion in exposure dwarfs the between-decile gradient, and they show that an optimal package retains a strictly positive price subsidy even when transfers can condition on income and past usage. That is a principled defence of a price-capping instrument, and it disciplines what this paper’s proportional-cushioning arithmetic can be taken to mean. Bonnet et al. (2025) reach the opposite conclusion for transport fuel in France, and the disagreement is commodity-driven, fuel budget shares rise with income where domestic-

energy shares fall, which is precisely the kind of composition effect the present framework is built to make visible. Auclert et al. (2023) add the externality a domestic incidence exercise cannot see: national price subsidies insulate one importer at the cost of world energy demand. Cross-country, Ari et al. (2022) and Amaglobeli et al. (2026) document that most European crisis support was untargeted and price-distorting, while NAO (2024) records the fiscal counterpart in the UK: an initial £139 billion estimate for the eight energy support schemes against a £44 billion outturn, of which the Guarantee itself was £24.0 billion, the option value of support that binds only in bad states.

2.4 The UK episode record

Each episode carries its own first-stage literature, reviewed with the imported numbers in Section 3.7: the Brexit/TCA price and trade effects (Bakker et al., 2023, 2026; Breinlich et al., 2022; Freeman et al., 2022), the 2022 terms-of-trade and energy-price shock and its policy response, the 2025 US tariff schedule (OBR, 2025), and the India agreement’s impact assessment (DBT, 2025). The Brexit episode has the richest distributional record, and two precedents sit close to this paper. Breinlich et al. (2016) asked literally who would bear the pain, distributing projected Brexit price changes across income groups and household types, and Clarke et al. (2017) did the same for an MFN tariff scenario, finding the largest hits on unemployed, single-parent and pensioner households. Both are ex-ante scenario exercises predating the actual border; this paper’s contribution relative to them is realised episodes, a statutory second stage, and comparison across shocks. Davenport and Levell (2021) supply the earnings-side counterpart (barrier exposure by occupation, sector and region), and Alabrese et al. (2026) and Dhingra et al. (2026) the 2026 regional accounting, the latter with new TCA tariff and non-tariff-measure estimates in a spatial general-equilibrium model, finding the damage runs through real wages and productivity rather than structural change.

A near-namesake deserves flagging: Pallotti et al. (2023) ask who bore the costs of the euro-area inflation shock and find consumption losses roughly flat across the income distribution, with the incidence falling instead along an age gradient as rigid rents hedged lower-income households. That the same question yields a vertical answer in Britain and an age-based one in the euro area is itself a reason to hold the lens fixed and vary the shock.

The record also contains a puzzle this paper is built to address. Breinlich et al. (2022) find the depreciation episode’s consumer cost roughly *flat* across the income distribution; Bakker et al. (2026) find the non-tariff-barrier episode sharply *regressive*, the bottom decile’s cost-of-living increase around half again larger than the top’s. Same country, adjacent years, opposite incidence, because the two shocks landed on different baskets. A framework that holds the household lens fixed and varies the shock is the instrument that settles such disagreements, and Section 4 shows the same mechanism operating across five episodes. What remains absent everywhere is the tax–benefit response: the Brexit price studies reviewed here do not model the tax–benefit response; the energy-crisis literature scores the discretionary response but not the trade shock as such; the FTA impact assessments contain, in DBT’s own regulator’s words, household analysis “developed further for future FTA IAs.” The gap this paper fills, for the two price-led episodes, is the join: one statutory second stage run against the episode’s first stage. The paper supplies the first stages, the taxonomy, a public-data first pass for all five episodes, and the statutory stage for the two largest; extending that stage to the remaining three is what would complete the comparative argument.

3 Framework, episodes and method

3.1 The estimand

For every episode the object is the same conditional quantity: the statutory household response to a declared trade-shock first stage. Let a first stage specify (i) a vector of consumer-price changes by consumption category, and/or (ii) a labour-income loss or gain with a sectoral location and an adjustment margin. The second stage maps that first stage onto households and computes the change in disposable income, its distribution, and the Exchequer’s share, under the tax–benefit rules in force at the episode date. The headline metric is the *cushioning rate*. For an earnings shock it is the income-stabilisation coefficient of [Dolls et al. \(2012\)](#),

$$c = 1 - \frac{\Delta Y^{\text{disp}}}{\Delta Y^{\text{mkt}}},$$

one minus the disposable-income change per pound of market-income shock. For a price shock, disposable income does not move (the structural-zero result of Section 5), so the generalisation must be stated rather than gestured at:

$$c = \frac{T}{\Delta E}, \quad \Delta E = \sum_k p_k q_k \hat{p}_k,$$

where ΔE is the expenditure-weighted cost of the price vector $\hat{p}$ at fixed consumption and T is the sum of the statutory transfers the episode triggered (automatic plus discretionary). The two coefficients answer the same question, what share of the gross shock the state returned, on the margin the shock arrived on. Ratios, unlike levels, are comparable across episodes whose gross sizes span two orders of magnitude.

Nothing in the design estimates the first stages. Each is imported from published estimates and labelled with its source and its attribution caveats (Section 3.7); the paper’s contribution begins where those estimates stop, at the household and the Exchequer. This is the conditional-estimand convention of the companion tariff study ([Ahmadi, 2026](#)), applied retrospectively, with the advantage that retrospective first stages can be anchored to realised outturns rather than ex-ante bridges.

3.2 Two channels

Trade shocks reach households through prices and through earnings, and the post-Covid episodes split cleanly on that line. The consumption channel prices a category-level price vector against household expenditure shares: implemented at the household level on imputed consumption, done for the energy episode in Section 5, and, for the other episodes, on published expenditure-by-decile tables. The earnings channel imposes sectoral labour-income changes on employees by industry and runs them through the statutory system, with the adjustment margin (wage cuts versus displacement) as a declared scenario dimension; the machinery, conventions and caveats of the companion study carry over unchanged. The channels are reported separately and never summed without an explicit interaction note: a price shock erodes real income that the benefit system partially indexes, while an earnings shock triggers means-tested entitlement; the two responses run through different statutory mechanisms, and their comparison is the point.

3.3 The rulebook axis

The same shock cushions differently depending on when it lands, because the rulebook moves: benefit uprating follows lagged inflation, so a fast price shock outruns indexation for a year; thresholds frozen in cash terms tighten in real terms through an inflation episode; and discretionary interventions (most dramatically the Energy Price Guarantee) rewrite the effective schedule mid-shock. The design therefore separates three quantities for each episode: the cushioning delivered by the rules *as they stood*, the cushioning the same rules would have delivered under contemporaneous full indexation (the uprating-lag cost), and the discretionary layer on top. Price-shock incidence has been run through tax-benefit models before (Section 2), and indexation counterfactuals are standard in that literature; what this paper adds is to hold a *trade* shock fixed and vary the rulebook vintage that receives it, and to separate the statutory timing lag from the discretionary layer within a single episode. The post-Covid sequence, four non-benchmark episodes landing on four different vintages of the same welfare state, is a natural laboratory for exactly that comparison.

3.4 Identification: where the burden sits, and where it does not

Because the paper reports household effects of named policy episodes, it is worth stating precisely which of its numbers require an identifying assumption and which do not. The design deliberately separates the two.

The second stage requires no identification. Given a price vector and a household’s observed characteristics, what the tax-benefit system pays that household is a computation on statutory rules, not an estimate of a causal parameter. That the means-tested system returns nothing when nominal incomes do not move, that the Energy Price Guarantee returns the difference between two capped prices, that a flat payment is worth more relative to a small bill, these are properties of the rulebook, verifiable line by line, with no counterfactual inference involved. The same is true of the variance decomposition: it describes the dispersion of a computed quantity across observed households. This is the sense in which the exercise is a calculator rather than an estimator, and it is why the second stage can be run at all.

The first stages carry all the identification, and it is borrowed. Whether the Trade and Cooperation Agreement *caused* the food-price rise is identified, or not, by Bakker et al. (2023), whose cross-product design and its limits the paper inherits wholesale. Whether the 2025 tariffs will cause the earnings loss the companion study imposes is not identified by anyone; it is declared. The government’s projections for the India agreement are model output, not estimates. The design’s purpose is to quarantine this problem: each first stage is labelled with its epistemic status (Section 3.7), and no result in this paper is stronger than the first stage it conditions on.

One attribution is accounting, not inference. The claim that part of the 2022 energy shock is trade-transmitted rests on Ofgem’s published decomposition of the price cap into wholesale and non-wholesale components, a statement about the composition of a regulated price, not a causal estimate of what world gas markets did to British households.

What follows is that the paper’s results should be read as *conditional statutory incidence*: given that a shock of a stated size and shape occurred, this is how the rulebook distributed it. They do not identify the causal effect of Brexit, the tariffs or the agreements on household welfare, and any sentence in this paper that reads that way should be read as shorthand for the conditional statement. The title’s question is answered in that sense: not who was harmed by trade policy, but who bore a declared shock once the state had finished responding.

3.5 What a decile lens can and cannot see

Two recent results bound what any distribution-by-decile exercise, including this paper’s first pass, is entitled to claim. [Borusyak and Jaravel \(2024\)](#) show that for trade shocks the overwhelming share of the variance of household welfare changes lies *within* income deciles rather than between them, because exposure is driven by product-level consumption patterns that income predicts weakly; [Levell et al. \(2025\)](#) reach the same conclusion for energy on UK transaction data. The implication is not that decile gradients are wrong (they are real, and they are what Section 4 measures) but that they are a small part of the dispersion in who was hurt. Decile means answer “was this shock regressive?”; they do not answer “who was hurt?”, and the two questions have different answers. This is the strongest single argument for the household-record stage of Section 5, where imputed exposure and simulated entitlement are joined at the household level, and it is why the first-pass results are reported as gradients rather than as welfare statements.

A second bound concerns channels. Decomposing inflation’s household impact, [Ferreira et al. \(2026\)](#) find the net-nominal-position and labour-income channels an order of magnitude larger than the consumption-basket channel. The basket channel is the entire content of the price-side exercise here: savers and borrowers, mortgagors and depositors, are not in this paper’s ledger, and the earnings channel enters only for the one episode where the companion study supplies it. The estimates are therefore a lower bound on the distributional consequences of these episodes, and are labelled as such throughout.

3.6 What the design is not

The estimates are first-round statutory incidence, not general equilibrium: no behavioural response, no wage or employment feedback from the price shocks, no substitution in consumption (the price-channel numbers are first-order, envelope-theorem approximations to welfare changes and upper bounds on avoidable cost). Episode first stages carry their own literatures’ uncertainty, which the design brackets by importing low/central/high variants where the literature provides them and labelling single-point stages as such. The comparison across episodes is a comparison of conditional scenarios that share a second stage, not a causal decomposition of realised household outcomes over 2021–2026, which would require separating the trade shocks from everything else that happened, a task this design deliberately declines.

3.7 The five episodes and their declared first stages

Each episode enters the analysis as a declared first stage: published estimates, imported with citations and attribution caveats, never re-estimated. The five span the design’s axes: negative and positive, price-led and earnings-led, measured and projected, policy-made and trade-transmitted. One is carried as a near-zero benchmark. Figure 1 places the set on the paper’s two axes and Table 1 lists each declared first stage with its source and epistemic status; the evaluation-practice observation the dossier surfaces, that the sources reviewed here include no published UK household-level ex-post evaluation of a trade agreement, is developed in Section 6.

3.7.1 Episode 1: the TCA border (2021–23), negative, price-led

The Trade and Cooperation Agreement took effect in January 2021, replacing single-market membership with a zero-tariff but non-tariff-barrier-laden border. The price evidence is the



Figure 1: The episode set. Horizontal: the channel through which the shock reaches households; vertical: cost or gain. Bubble area is log-scaled in the gross shock. The 2022 energy episode (hatched, dashed ring) is trade-transmitted rather than a policy act; CPTPP (open, dashed) is the near-zero benchmark. Positions are conceptual classifications by sign and channel, not measured coordinates.

Table 1: Declared first stages. Every price or earnings vector is an imported published estimate; none is re-estimated in this paper.

Episode	Date	Type	Channel	First-stage source	Key imported number	Status
E1 TCA food NTBs	Dec 2019–Mar 2023 ^w	Policy	Consumer prices (food)	Bakker et al. (2023)	+8% on food prices (6% low variant)	Estimate (ex post)
E2 Energy 2022–23	Apr 2022–Mar 2023	Market	Consumer prices (energy)	Ofgem (2021, 2022) ; OBR (2023)	Cap £1,277 → £3,549; EPG £2,500	Observed (statutory)
E3 US tariffs 2025	2025	Policy (foreign)	Earnings only; no UK retaliation	Ahmadi (2026)	£886m/yr gross earnings shock	Scenario (unit stress; unpublished)
E4 India CETA	2026+	Policy	Consumer prices (clothing, footwear, food)	DBT (2025)	£180m/yr duty cut on final goods	Projection (ex ante)
E5 CPTPP benchmark	Long run	Policy	Aggregate GDP; no household mapping	DBT (2023)	+£2.0bn GDP (+0.08%)	Projection (long run)

Notes. Type distinguishes a policy instrument from a trade-transmitted market price. Source entries are short cite keys; see the bibliography. E3’s consumer-price row is exactly zero (the UK imposed no retaliatory tariffs), so its household incidence runs through earnings alone. ^w the *estimation* window of the cited study; the TCA itself entered into force in January 2021.

cleanest of any episode: exploiting cross-product variation in EU import intensity, [Bakker et al. \(2023\)](#) attribute roughly 8 percentage points of the 25 per cent food price rise between December 2019 and March 2023 to Brexit non-tariff barriers, a cumulative £250 per household on food alone, £6.95 billion in aggregate (their household count; £250 on the ONS count used here gives £7.1 billion), with pass-through of NTB costs to consumers of 50–80 per cent and the largest effects in high-EU-import products such as meat and dairy (around 10 points above low-exposure comparators). The published 2026 version of the same study revises the estimate upward, to a 4 per cent rise a year and 12 per cent over three years, with household losses of \$14.8 billion ([Bakker et al., 2026](#)); the declared first stage stays on the 2023 vintage whose £250 cumulative anchors the pipeline, so the food-episode level should be read as the lower of the two published estimates. The trade-volume record: relative UK imports from the EU fell about 25 per cent on entry ([Freeman et al., 2022](#)), and in 2022 goods exports were an estimated £27 billion (6.4 per cent) below counterfactual, with some 16,400 firms exiting EU exporting entirely ([Freeman et al., 2024](#)). The earnings side, by contrast, rests on assumptions rather than estimates (the OBR’s standing 15 per cent trade-intensity and 4 per cent productivity wedges, [OBR, 2026](#)), so this episode’s earnings channel is declared as an assumption-based scenario and its price channel as an estimate-based one. Attribution caveat: the price identification is relative (cross-product), so economy-wide effects may be missed, and the counterfactual is contested by critics of the OBR’s trade assumptions.

3.7.2 Episode 2: the terms-of-trade shock (2022–23), negative, price-led, very large

The 2022 energy and import-price surge was, in the Bank of England’s description, a terms-of-trade shock of a size unseen since the mid-1970s: the terms of trade deteriorated 6.2 per cent in the year to 2022Q3 (ONS, 2023), import prices rose 20 per cent relative to output prices over two years (Broadbent, 2022). The energy import bill roughly doubled to £117 billion in 2022 (Offshore Energies UK, 2023), the *increase* of roughly £60 billion being a transfer abroad of about 2.5 per cent of GDP (derived arithmetic, labelled as such); of the roughly 17 per cent CPI rise to 2022Q4, more than 70 per cent (12.7 points) was imported (Dhingra and Page, 2023). The household-level price vector is policy-mediated and well documented: the typical annual energy bill’s cap rose from about £1,277 (winter 2021–22) to £3,549 (October 2022), truncated to £2,500 by the Energy Price Guarantee, at a discretionary cost the OBR put at £27 billion for the EPG within £47 billion of price subsidies and £78 billion of total energy support (OBR, 2023). Counterfactual no-policy welfare losses average 6 per cent of income, with some households facing much larger losses (Levell et al., 2025), against energy budget shares of roughly 11 per cent in the second income decile versus 6 per cent in the ninth (Resolution Foundation, 2022). This is the strongest first stage of the five, multiple independent official estimates of both the aggregate shock and the retail price vector, plus a published imported/domestic decomposition, and the design decision it forces is constitutive for this paper: the shock is declared *gross*, and the Energy Price Guarantee is modelled as what it was, a discretionary rulebook intervention, so that automatic and discretionary cushioning can be separated.

3.7.3 Episode 3: the US tariffs (2025–26), negative, earnings-led

The spring 2025 US schedule (a 10 per cent baseline and 25 per cent rates on autos and on steel and aluminium, the latter from March) and its partial mitigation are the subject of the companion study (Ahmadi, 2026), which supplies this episode’s earnings first stage and its second-stage results: a tariff-calibrated earnings loss of roughly £886 million a year, cushioned at 43.9 per cent when delivered as broad wage cuts and 36.6 per cent as displacement. The external anchors are the OBR’s March 2025 scenarios (import-demand elasticity 0.4; peak GDP effects of -0.6 to -1.0 per cent across scenarios) (OBR, 2025) and the Bank of England’s estimate of an effective US tariff on UK goods of about 11 per cent at April 2025 (Bank of England, 2025). No UK retaliatory tariff schedule had been imposed as of August 2026, so the direct consumer-price channel of this episode is set to zero; indirect channels (exchange rates, trade diversion, the Economic Prosperity Deal’s own tariff changes) are outside the design.

3.7.4 Episode 4: the India agreement (2026–), positive, both channels

The UK–India Comprehensive Economic and Trade Agreement entered into force on 15 July 2026. Its first stage is declared from the government’s own impact assessment (DBT, 2025): long-run GDP $+\text{£}4.8$ billion a year ($+0.13$ per cent by 2040), real wages $+0.19$ per cent ($\approx \text{£}2.2$ billion), and, on a different clock, after tariff staging of up to ten years, consumer duty reductions of £220 million a year, £180 million of it on final goods, concentrated in clothing (£132 million), footwear (£13 million) and food and beverages (£12 million). Sectoral GVA effects give the earnings channel its geography: machinery and equipment $+1.65$ per cent, textiles and apparel -0.68 per cent. Two dials are declared rather than estimated, following the IA’s own caveats: preference utilisation (the duty figures assume 100 per cent on static 2022 trade) and consumer pass-through of the duty savings. The framing, and the reason a government projection can

anchor a conditional design, is that the exercise computes what the government’s own central estimates imply for households, which its impact assessment does not report and its regulatory scrutineer asked for (RPC, 2025).

3.7.5 Episode 5: CPTPP accession (2024–), the near-zero benchmark

The UK’s accession to the CPTPP (in force December 2024) carries a central government estimate of +£2.0 billion a year in the long run, 0.08 per cent of GDP, about £70 per household per year before any incidence structure, because the UK already had agreements with nine of the eleven members (DBT, 2023). It is included precisely because it is nearly nothing: a data-limitation benchmark (the replication artifacts label it accordingly) whose only defensible treatment, an aggregate divided by a household count and nothing more, marks the boundary of what a declared first stage can support.

3.8 First-pass method

This section describes the first pass, which runs the declared first stages against public aggregates for all five episodes; the energy episode is additionally run on household records in Section 5, whose method is set out there. Everything is generated by a pinned pipeline that verifies its raw inputs by hash and emits every reported value as a machine-generated macro or generated table; the declared first-stage constants of Section 3.7 are typed *with their citations* and marked as imported, and a small number of external anchors and convention settings (cap levels, scheme amounts, sensitivity dials) appear as literals beside the macros they condition.

Following the companion studies’ convention, the epistemic status of every input is stated once:

Object	Status
Episode first stages	Imported from cited published estimates (episodes 1–2) or government projections (episodes 4–5); episode 3 from the companion study. Never re-estimated here.
Expenditure by category × decile	Data (ONS Family Spending detailed workbooks).
Energy price-cap levels, statutory benefit rates, CPI outturns	Data (Ofgem/DESNZ, DWP, ONS).
Pass-through and utilisation dials	Assumed , bracketed, entering the pipeline once each.
Incidence arithmetic	Computed ; no estimation step anywhere in the pipeline.

Consumption incidence. Each episode’s price vector is applied to the relevant categories of household expenditure by equivalised-disposable-income decile, giving a per-decile burden or gain in pounds a year, as a share of total spending, and as a share of disposable income. These are first-order calculations: no substitution, no behavioural response, fixed baskets, upper bounds on avoidable welfare cost, in the envelope-theorem sense of Deaton (1989), whose first-order incidence method this literature, from Fajgelbaum and Khandelwal (2016) onward, descends from. Where a published aggregate exists, its role differs by episode and is stated in the results: the OBR’s Energy Price Guarantee cost is a context anchor beside the pipeline’s own aggregate, while the TCA’s £250 published cumulative is used as an *anchor*: the pipeline’s linear-ramp

cumulative overshoots it by a factor of 1.68, so the headline TCA path is rescaled to the published figure (Section 4), so that episode’s level is the published estimate’s, not the pipeline’s.

Rulebook arithmetic. The vintage axis is seeded with statutory arithmetic on public parameters: the real-terms shortfall of the Universal Credit standard allowance during the 2022–23 inflation episode relative to contemporaneous indexation (the uprating-lag cost, computed month by month against CPI outturns), and the discrete removal of the £20-a-week uplift in October 2021, the rulebook event that immediately preceded the terms-of-trade shock. The Energy Price Guarantee enters as the discretionary layer of episode 2: the gap between the gross price shock and the EPG-truncated shock is the discretionary cushioning share, computable per decile from cap levels alone.

Two data caveats. The FYE2025 Family Spending workbook carries an ONS correction notice affecting percentage standard errors across the high-level COICOP categories; the point estimates this paper uses are unaffected, but no sampling-uncertainty statement is made on the expenditure inputs for that reason. And the two vintages used (FYE2022 for the TCA and energy episodes, whose spending bases must predate the shocks, and FYE2025 for the India agreement) sit either side of an ONS change in the equivalence scale, so income levels are not comparable across vintages, which is why every cross-episode comparison in this paper runs on shares of spending rather than shares of income.

Comparability. Episodes are compared in ratios, not levels: burden as a share of spending by decile and cushioning shares where computable, because gross sizes span more than two orders of magnitude and pound totals would reduce the comparison to a statement about the energy episode. Scenario dials (pass-through, utilisation) are reported as labelled variants, never blended into central cells; brackets are outer bounds, not confidence intervals; and estimate-based versus projection-based first stages are flagged in every table.

4 First-pass results

4.1 The comparative anatomy

Table 2 places the five episodes side by side under the common lens, and Figure 2 plots the three price profiles it summarises. First, *the negative shocks were regressive and the positive episodes were not progressive*: on the realised 2022–23 path the energy shock cost the bottom decile 5.93 per cent of its spending against 2.76 per cent at the top (the never-realised gross counterfactual would have been 8.95 against 4.17), the TCA food shock 0.35 against 0.20 per cent of spending on the window-average path (1.19 against 0.66 at end-state), while the India agreement’s consumer gains are spread almost exactly proportionally (under 0.02 per cent of spending in every decile) and are two orders of magnitude smaller than the energy shock and one order smaller than the food shock. Second, *regressivity follows the affected category’s budget-share gradient*: food takes 14.9 per cent of bottom-decile spending against 8.3 at the top, energy 7.0 against 3.2, while clothing and footwear, where the India CETA gains land, are flat across the distribution: the composition of a shock, not just its size, determines who bears it. Third, *the modelled responses run through different mechanisms*: the one earnings-led episode, a scenario imported from the

unpublished companion study, is cushioned through the automatic tax–benefit system (36.6–43.9 per cent), while the price-led energy episode was cushioned by discretionary instruments: the Energy Price Guarantee absorbed 33.8 per cent of the gross shock on the financial-year basis (the point-to-point cap ratio is 46.2 per cent), and the automatic price-side mechanism, benefit indexation, arrived a year late (Section 4.3).

Table 2: Five shocks under one lens: first-pass comparability

Episode	Type, channel	Shock (realised path)	% of spending		Cushioning, by mechanism
			D1	D10	
TCA food, 2021–23	policy, prices	£2.2 bn	0.35	0.20	none legislated (0%) ^t
Energy, 2022–23	market, prices	£30.6 bn	5.93	2.76	price subsidy, 33.8% ⁱ
US tariffs, 2025–26	policy, earnings	£886 m	zero		— ^m
India CETA, 2026–	policy, prices	£180 m ^g	0.0151	0.0154	—
CPTPP, 2024–	benchmark	£2.0 bn ^g	no structure		—

Notes: ^g gain, not cost. ^t the common cushioning estimand, computed at the household stage (Section 5.7): no discretionary instrument was legislated against this episode, and no modelled entitlement reads food consumption, so the cash-benefit response is zero by construction. ⁱ the Guarantee alone, as a share of the *gross* counterfactual shock (£46.2 bn on the financial-year basis); the realised shock in the size column is net of it. Because the Guarantee capped a price, the share is constant across deciles by construction (the point-to-point cap ratio is 46.2 per cent); with the Bills Support Scheme the energy-specific stack returns 59.5 per cent at the household stage (Section 5). ^m this column reports the cushioning estimand of Section 3 for the two price episodes; the tariff episode’s earnings-channel stabilisation coefficient (36.6–43.9 per cent, imported from the companion study (Ahmadi, 2026)) answers a different question and is reported in this note rather than in the column. Deciles are of equivalised disposable income; entries are first-order incidence on published expenditure-by-decile tables at fixed baskets. Both price episodes are on their *realised* paths, but on different clocks, and the asymmetry matters for the D1 comparison: energy is an annualised crisis-year rate, while TCA is the published cumulative averaged over a window padded with pre-entry zeros, at the end-state (crisis-period) rate the TCA row reads 1.19 against 0.66 per cent. Energy is on the caps households actually faced in 2022–23 (£2,236 against £1,208, +85.1 per cent). The end-state and gross-counterfactual variants are reported in the text. Episodes are typed policy/market: the energy episode is trade-*transmitted*, not a policy act. “Zero” records that the UK did not retaliate, so no consumer price vector exists. Cushioning entries measure different mechanisms (a pass-through convention, a price subsidy, a tax–benefit stabilisation coefficient) and are not summable. Category budget-share gradients, which generate these incidence patterns, are in Figure 3; full per-decile detail in Table 3 (whose derived columns are computed by the figure generator from the pinned artifact, on the same FY basis).

4.2 Episode detail

TCA food prices. The headline incidence uses the window-average path implied by the published cumulative: the £250-per-household estimate implies an average effect of about 30 per cent of the 8 per cent end-state over the window, giving £58 (bottom decile) to £94 (top decile) a year, £2.2 billion a year in aggregate, regressive as a share of spending (0.35 versus 0.20 per cent) because food takes 14.9 per cent of bottom-decile spending against 8.3 at the top. (The estimation window opens in December 2019, thirteen months before the TCA took effect; the linear-ramp and window-average conventions both absorb the pre-entry months as zeros.) This is an anchoring choice: the pipeline’s own linear-ramp cumulative overshoots the published figure by a factor of 1.68, and the window-average path is obtained by rescaling to the published cumulative. The TCA row therefore carries no information from the pipeline beyond the ONS food-share gradient, and the level is the published estimate’s. The end-state figures

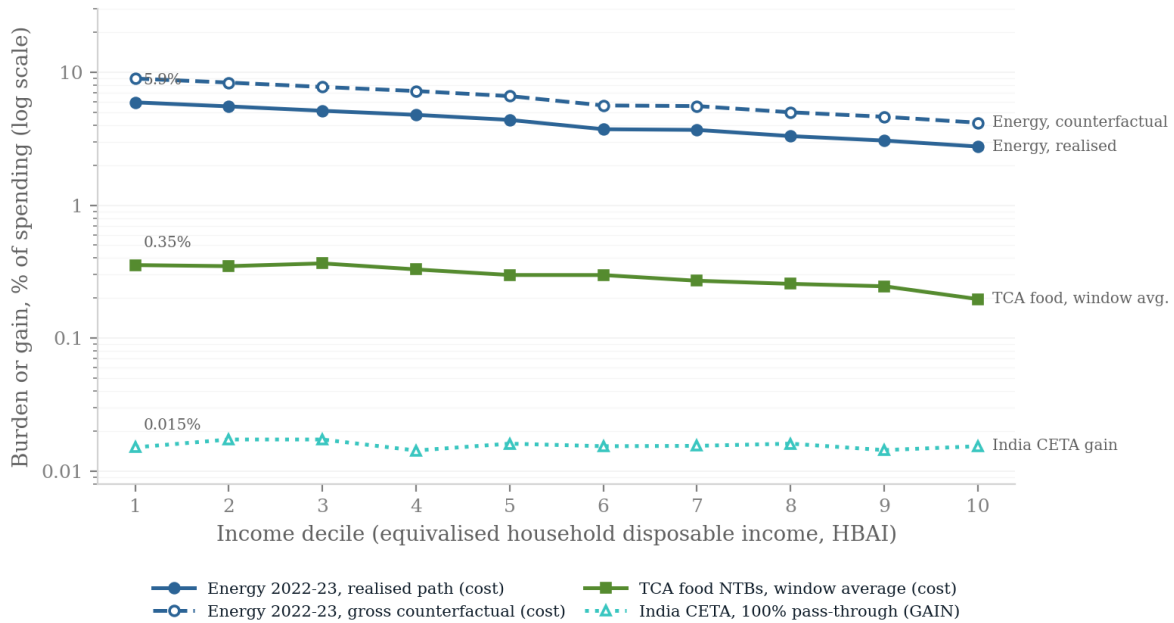


Figure 2: Household incidence by equivalised disposable income decile, as a share of each decile's total spending. All series are plotted as magnitudes on a log axis, so the chart compares sizes, not signs. Costs are plotted as positive; the India CETA series is a gain (dotted, open markers). The energy episode is shown on its realised 2022–23 path (solid) and as the gross counterfactual the Energy Price Guarantee displaced (dashed).

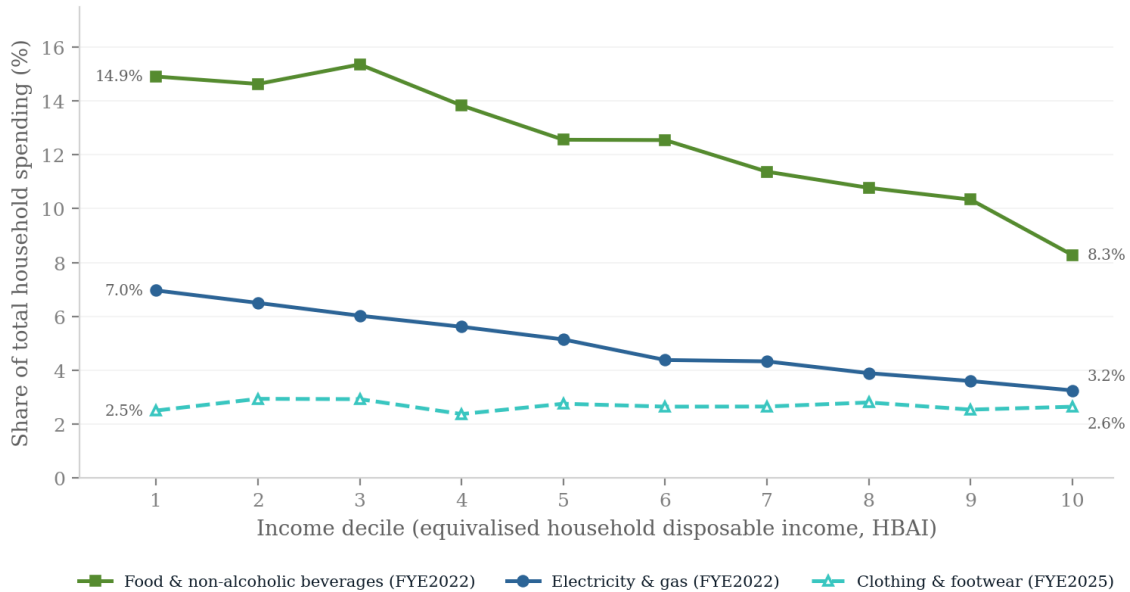


Figure 3: Why each episode has the incidence it does: the affected categories' shares of total spending by decile. Food and domestic energy fall steeply with income; clothing and footwear, where the India CETA gains land, are flat.

Table 3: Household incidence of the quantified episodes, by equivalised disposable income decile. TCA food NTBs are on the window-average path (0.30 of the end-state +8% effect); energy is the 2022–23 cap shock, gross and net of the Energy Price Guarantee; the India CETA column is the consumer gain at 100% pass-through. Costs are positive.

Decile	TCA food NTBs		Energy, gross		Energy, net of EPG		CETA gain
	£/yr	% spend	£/yr	% spend	£/yr	% spend	£/yr
1	58	0.35	1,458	8.95	965	5.93	2.96
2	63	0.35	1,504	8.35	996	5.53	3.93
3	70	0.37	1,478	7.74	978	5.12	4.47
4	74	0.33	1,631	7.21	1,080	4.78	3.90
5	72	0.30	1,591	6.61	1,054	4.38	5.12
6	84	0.30	1,584	5.62	1,049	3.72	5.37
7	80	0.27	1,651	5.56	1,094	3.68	5.61
8	84	0.26	1,631	4.99	1,080	3.31	6.56
9	90	0.25	1,698	4.62	1,124	3.06	6.90
10	94	0.20	1,986	4.17	1,315	2.76	9.93
D1/D10 ratio	—	1.80	—	2.15	—	2.15	0.98 [†]

Notes. £ per household per year; % spend is the burden or gain as a percentage of that decile’s total expenditure. The EPG cushions 33.8% of the gross energy shock at every decile by construction. Ratios are computed on unrounded values, so they may differ from the displayed cells in the last digit. [†] CETA ratio is on the %-of-spending basis (£ gains are not comparable across deciles). Spending bases: ONS Family Spending workbook 1, sheet 3.1E, FYE2022 vintage (TCA, energy) and FYE2025 vintage (CETA).

(£194–315 a year; £7.37 billion) are the unrescaled variant; the 1.68 overshoot shows the price effect was back-loaded, so end-state overstates the lived burden. The household stage would carry the published path if this episode were extended to it (Section 6.1).

The energy shock and the EPG. What households actually faced in 2022–23 was a mean capped bill of £2,236 (£1,971 for the first half-year, the Energy Price Guarantee’s £2,500 for the second) against £1,208 the year before: a realised rise of 85.1 per cent costing the bottom decile £965 a year, 5.93 per cent of its total spending against 2.76 per cent at the top, and £30.6 billion in aggregate. The *counterfactual* shock, the October 2022 cap of £3,549 carried through the financial-year mean, giving £2,760 against the FY2021–22 mean of £1,208, +128.6 per cent, is the figure the policy prevented, and would have cost the bottom decile £1,458 a year, 8.95 per cent of its spending. Three labelled qualifications bound that number, and each is a different kind of bound. First, *welfare versus cash*: the fixed-basket figure is the first-order welfare measure, correct to a first approximation by the envelope theorem, but households cut consumption, so cash outlays rose by less. Applying the observed weather-adjusted reduction in domestic gas consumption of 13 per cent over the gas year to mid-May 2023 (DESNZ, 2024) gives a cash-outlay burden of 77 per cent of the fixed-basket figure (the gas-only response is applied to the combined electricity and gas bill; electricity demand fell by less, so the scale slightly overstates the outlay reduction) (£1,128 for the bottom decile; £35.8 billion aggregate). The two answer different questions, what households paid, and what they lost, and the gap between them is itself the foregone consumption. Second, *how much of this is trade*: the retail cap bundles wholesale energy, which is traded, with network, policy and operating costs, margin and VAT, which are domestic. Ofgem’s published component breakdown settles the split (Ofgem, 2021, 2022).

Wholesale rose from £550 to £2,468 of the typical bill, from 43.1 to 69.5 per cent of the cap, so 84.4 per cent of the cap’s *rise* is traded energy and the remainder domestic cost. The trade-transmitted component of the realised burden is therefore about £25.8 billion (£39.0 billion of the gross counterfactual). One clarification matters for what this number means. Wholesale is not the same as imported: the wholesale price is set on a world-linked market, so UK-produced gas repriced too, and the resulting transfer runs from British households to British producers and, through the Energy Profits Levy, the Exchequer, not abroad. The wholesale share therefore measures exposure to a world-market-linked price rather than a transfer overseas, and the trade-transmitted figures above should be read in that narrower sense. The comparison with [Dhingra and Page \(2023\)](#)’s finding that *more than* 70 per cent of the CPI rise was imported is offered only as an order-of-magnitude check, since their statistic concerns the aggregate price index rather than the wholesale component of one regulated tariff. Third, *timing*: the counterfactual holds the October 2022 cap for a full year, which no household experienced. Because both paths are financial-year means, the £15.6 billion of Guarantee absorption already represents the instrument’s six months at annual spending; the OBR’s £27 billion lifetime costing (March 2023) is a context anchor on different prices and a longer window. Within those bounds the shape is robust: the Energy Price Guarantee truncated the rise to +85.1 per cent, absorbing £492 a year for a bottom-decile household, 33.8 per cent of the gross shock on the financial-year basis (£16.5 of £48.7 billion, Section 5); the point-to-point cap ratio, $(3,549-2,500)/(3,549-1,277)$, is 46.2 per cent and is not used further. Because the EPG capped the *price*, its cushioning share is constant across deciles by construction. Two qualifications follow from the frontier evidence. First, decile means understate the problem the instrument faced: [Levell et al. \(2025\)](#) show that income explains only a small share of the variation in household energy spending, so exposure within a decile is far more dispersed than the gradient between deciles, an argument for the household-record stage, where that dispersion is observable, and against reading any decile-mean cushioning share as a statement about individual households. Second, [Levell et al. \(2025\)](#) find that an optimal package in their welfare model retains a strictly positive price subsidy even when transfers can condition on income and past usage, because income targets energy exposure poorly. The finding here is descriptive: the cap cushioned proportionally. Whether the overall discretionary response was progressive cannot be judged from the EPG alone: the Energy Bills Support Scheme’s flat £400, the means-tested cost-of-living payments and the council-tax rebate, jointly of the same order as the EPG and deliberately targeted, are not modelled in this first pass. Section 5 reports the household-record result: with the whole modelled stack, the cushioning rate falls with income.

One comparator needs reconciling. [Resolution Foundation \(2022\)](#) report energy budget shares of roughly 11 per cent in the second decile against 6 per cent in the ninth; the shares computed here are 6.5 and 3.6 per cent. The gap is definitional, not substantive: this paper measures electricity and gas only, as a share of *total spending*, on equivalised-income deciles of the FYE2022 vintage, whereas the wider comparator includes other domestic fuels and is expressed against income. Because the episode’s incidence is linear in this share, the difference scales the level of every energy number reported here, another reason the household-record stage, which computes the share directly rather than importing it, supersedes this pass.

The US tariffs. The UK consumer side is zero, no retaliation occurred, so this episode’s incidence is entirely earnings-led: a gross loss of £886 million a year cushioned by the automatic tax–benefit system at 36.6 per cent (displacement) to 43.9 per cent (wage cuts), the companion

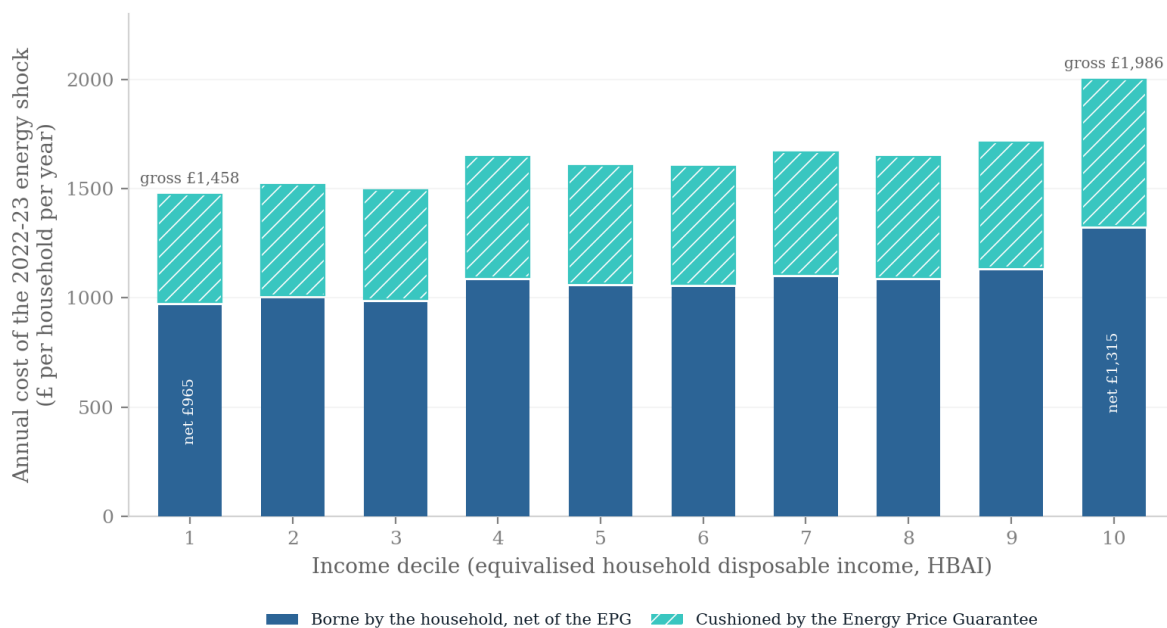


Figure 4: The 2022–23 energy shock by decile: the gross cap-level burden split into the part borne by households net of the Energy Price Guarantee and the part the Guarantee absorbed. Because the instrument capped a price, its cushioning share (33.8 per cent on the financial-year basis) is constant across the distribution by construction; the pound cushion runs from £492 in the bottom decile to about £670 in the top.

study’s headline carried over as this episode’s second stage.

The India gains, and the near-zero benchmark. Of the £180 million of final-goods duty reductions, £157.2 million falls in the three categories with published spending detail (clothing, footwear, food and beverages) and is allocated across deciles; the remainder is unmodelled. At full pass-through the CETA duty cuts are worth £2.96 a year to a bottom-decile household and £9.93 to a top-decile one, roughly proportional as a share of spending, though larger in pounds at the top (the top decile gains about three times as much in cash), and economically negligible at any pass-through dial (£1.48–4.97 at 50 per cent). The CPTPP row is a near-zero benchmark, included as a data-limitation case rather than a placebo test: its £2.0 billion long-run GDP estimate divides to £70 per household per year with no defensible distributional structure, and the pipeline reports exactly that and nothing more.

4.3 The rulebook seed results

The vintage axis is seeded with two statutory facts. First, the uprating lag: Universal Credit’s standard allowance rose 3.1 per cent in April 2022 (indexed to the previous September’s inflation) while prices rose at double digits. Against month-by-month contemporaneous indexation (an upper-bound counterfactual under which no actual benefit system operates), a single adult’s standard allowance fell £393 short over 2022–23 (9.8 per cent of the annual allowance); against the alternative of a single April 2022 uprating at contemporaneous annual CPI, £230. Both figures are for the single-adult standard allowance only, before elements and caseload weighting; the range, not either point, is the seed result. Neither the lag nor its cost is a discovery of this paper: [Cribb et al. \(2023\)](#) quantified the 2022–23 shortfall and scored the discretionary cost-of-living payments against a timely-indexation counterfactual, and [OBR \(2022b\)](#) put the aggregate real fall at roughly £12 billion, a figure the second stage’s £17.8 billion (Section 5) does not reproduce: the bases and price counterfactuals differ, the two are stated side by side rather than reconciled, and the reconciliation is an explicit task of the extension (Section 6.1). This paper’s addition is the join: the lag priced against trade-shock incidence, on the same households, across episodes. The lag also does not wash out over the cycle: [Judge and Murphy \(2023\)](#) count ten of fifteen years in which working-age benefits rose by less than outturn inflation. The shortfall arises from timing, and it coincides with the period in which the terms-of-trade shock peaked. Figure 5 shows the gap month by month. Second, sequencing: the £1,040-a-year Covid uplift was removed in October 2021, months before the energy shock landed, so the uplift’s end preceded the largest price-led trade shock of the period. Together with the EPG’s absorption of 33.8 per cent of the gross price shock, the seed picture is: an automatic response on the earnings margin (in the companion study’s scenario), a discretionary response on the price margin, and indexation that followed prices with a year’s lag, with the incidence of the full discretionary stack an open question for the microsimulation stage. The microsimulation stage turns these representative-household facts into population-weighted decompositions.

5 The second stage: the energy episode on households

5.1 Method

The largest episode is run through the statutory system on 53,508 enhanced-FRS household records in PolicyEngine UK 2.95.0 (the 2023–24 enhanced Family Resources Survey, 2023 vin-

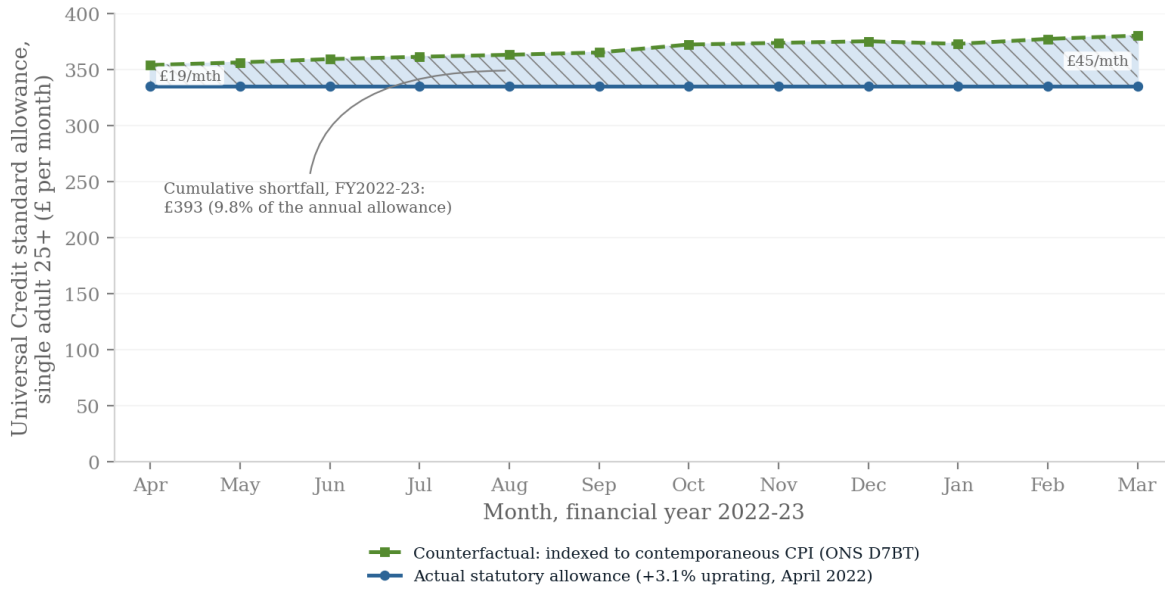


Figure 5: The uprating lag in the crisis year. The Universal Credit standard allowance for a single adult aged 25 or over (statutory path, flat from April 2022) against a counterfactual indexed to contemporaneous CPI. The shaded area is the shortfall, summing to £393 over 2022–23.

tag), in the tax–benefit microsimulation tradition surveyed by Figari et al. (2015). Four conventions govern the numbers that follow and each is a choice a reader could make differently.

One basis for both paths. The counterfactual and realised energy paths are both financial-year mean caps: the FY2022–23 counterfactual is £2,760 (the April and October 2022 caps, no Guarantee), the realised path £2,236 (April 2022 cap then the Guarantee), and the FY2021–22 baseline £1,208. This matters: comparing a point-to-point cap ratio with a financial-year mean, as is tempting, silently changes the denominator of the cushioning rate.

Exposure variable (declared). The household energy exposure is the model’s domestic energy consumption, whose mean (£1,992) exceeds electricity plus gas (£1,748) because it includes heating oil, LPG and solid fuels, which the cap did not cover. The mean divides out of every aggregate through the rebase, so the headline levels are unaffected; the distributional results treat off-grid fuel spending as if capped, a second-order approximation for the affected, disproportionately rural, households.

Rebasing (assumption). The enhanced FRS calibrates energy spending to current-vintage price levels, so its level reflects today’s prices while the cross-household distribution reflects quantities. The weighted mean is rebased to the FY2021–22 mean cap, a factor of 0.606, preserving each household’s relative position. Every shock aggregate below is linear in this factor and should be read as conditional on it; the within-decile results are invariant to it. The cushioning rates are not: the flat-cash instruments (the Bills Support Scheme, the cost-of-living payments) do not scale with the rebase while the shock denominator does, so the discretionary rate and the decile cushioning shares depend on the factor. The dependence is quantified in the two-axis grid of Table 4.

Coverage. The Energy Price Guarantee enters as the difference between the two capped

paths, the Energy Bills Support Scheme as its flat £400 to every weighted household (so the modelled £12.6 billion exceeds the £11.7 billion NAO outturn by about 7 per cent rather than the full 10.5 per cent weight excess; the model applies no supply screen, and the gap narrows because the outturn itself exceeds £400 times the ONS household count, the scheme having paid per electricity meter, second homes included), and the 2022–23 cost-of-living payments on simulated entitlement in the simulation year, an approximation in both directions: simulated entitlement is not receipt in the actual 2022 qualifying windows, the £150 disability payment is modelled per household rather than per qualifying individual, and Housing-Benefit-only households are included although they did not qualify for the £650. The council-tax rebate and the Warm Home Discount are not modelled. The discretionary total is therefore an approximation with omissions on both sides, not a bound.

Rule year (assumption). The simulation scores the episode on the 2023 parameter system and 2023 incomes, the vintage of the enhanced-FRS base, although the episode is FY2022–23. The rulebook-vintage run below scores the same households under the 2022 parameter system as the crisis-year counterfactual; the discretionary-stack results are insensitive to the rule year because the instruments enter as declared cash amounts, and the cost-of-living eligibility flags are nearly identical under the two vintages (8.1 against 8.2 million flagged households).

Joint error (declared). The disclosed vintage mismatches, 2023 incomes for a 2022–23 episode, an FYE2022 spending base, grossing weights 10.5 per cent above the ONS count, and the roughly five per cent first-pass gap, jointly imply an error class of roughly ± 10 per cent on every level in this section. Figures are reported to three digits for reproducibility against the pipeline, not as claimed accuracy.

Weights. All weighted arithmetic is delegated to the survey-weighting library that PolicyEngine itself returns results in; no weight calculation is hand-rolled. The person weights gross to 69.8 million people, in line with the UK population. PolicyEngine’s household grossing weights total 31.4 million households against an ONS count of 28.4 million, so aggregates carry an upward bias of about 10.5 per cent; ratios are unaffected. Deciles are equivalised on the HBAI before-housing-costs concept, and burdens are expressed as shares of household expenditure rather than income, because a ratio with income in the denominator is dominated by very low incomes. Both sides of every share sit on one base: the burden numerator is rebased to FY2021–22 cap levels, and the expenditure denominator to the ONS FYE2022 mean total-spending base the first pass divides by (factor 0.527), so second-stage shares are level-comparable with the first pass. The rebase factors are common scalars, so the variance decomposition and all quantile ratios are invariant to them (the within-share is 93.3 per cent on either base, verified by regeneration). Aggregate pounds, which share a basis, reconcile to within about five per cent.

Reconciliation with the first pass. On the same financial-year basis the first pass puts the counterfactual shock at £46.2 billion and the realised path at £30.6 billion, against £48.7 and £32.3 billion here. The residual gap of about five per cent is smaller than the 10.5 per cent excess of PolicyEngine’s grossing weights over the ONS household count, so the household base explains at most part of it: the imputed consumption distribution and the rebase convention offset the remainder, and the decomposition of the gap is not attempted here. Ratios are unaffected and levels should be read with the unreconciled gap in mind.

5.2 The automatic response is structural

Every means-tested award in the UK conditions on nominal income, and a price shock leaves nominal income untouched, so the fourteen cash benefits cannot respond within the year: the

zero is a property of the statutory rules and of the fixed nominal inputs, not an empirical finding. The tax-benefit calculator is run twice, once on baseline inputs and once with energy inputs shocked to the realised path, and the change across the fourteen benefits is £0.00 billion (0.0 per cent of the shock). The model does contain one entitlement that reads energy consumption: PolicyEngine UK’s own Energy Price Guarantee variable, which pays £14.9 billion in the 2023 baseline and rises by £1.8 billion under the shocked inputs, moving household benefits by the same amount. That delta is measured around the dataset’s current-vintage energy levels (the shocked inputs sit about twelve per cent above the unrebased baseline), so it verifies that the variable responds; it is not comparable to the £16.5 billion cushion, which is defined against the FY2021–22 base. The paper classifies the Guarantee as discretionary and models it on the financial-year basis of Section 5.1 rather than through the model’s calendar-year variable; the model’s cost-of-living payment variable (£10.4 billion at 2023–24 amounts) sits inside the HBAI income that ranks the deciles, as do the native Guarantee (£14.9 billion) and the £3.8 billion energy-bills rebate, so the deciles are ranked on income inclusive of roughly £29 billion of 2023 support instruments; the cushioning arithmetic itself counts each hand-rolled instrument once. It is also a first-order claim: a shock of this size moved nominal incomes in general equilibrium, wage bargaining, self-employment margins, and those excluded channels (Section 3) would trigger means-tested responses the first-order calculation cannot. This is [Cohen and Follette \(2000\)](#)’s macro finding, that automatic stabilisers do little against a supply shock, at the household level: for a pure price shock with nominal incomes fixed, the means-tested response is zero.

Two qualifications bound the claim. It is a statement about the *modelled* statutory system, so channels outside it, the Warm Home Discount, capital drawdown altering means-tested entitlement, passported eligibility shifting at the margin, self-employed income responses, could move the true figure off zero, though all are small relative to the shock. And it is an in-year statement: indexation is the system’s automatic price-side channel, and it arrives late rather than never.

5.3 Indexation arrives late

Benefits rose 3.1 per cent in April 2022, indexed to the previous September, against inflation over the year of about 10 per cent, the benefit-erosion mechanism whose cross-national treatment is [Paulus et al. \(2020\)](#). The cost of that gap is here computed with a genuine rulebook-vintage run rather than a flat base-times-gap multiplication: the enhanced FRS is a single-period dataset, so every input column is copied to the 2022 period and the *same* households are scored under the 2022 and 2023 parameter systems. Scoring the same households under the 2023 parameter system instead of the 2022 system moves the fourteen cash benefits from £233.2 to £254.6 billion: £21.3 billion through the full statutory structure, tapers included. The April 2023 uprating (10.1 per cent, applied to thirteen of the fourteen; the winter fuel payment is a fixed cash amount) is the largest change between the two systems but not the only one: the benefit cap and Universal Credit work allowances rose under the same order, the Universal Credit childcare caps rose in June 2023, and the National Insurance primary threshold rose in July 2022, so £2.11 billion per uprating point is a calibration that a parameters-only reform would refine. At that rate, the 7.0 points by which the April 2022 uprating fell short of the September 2022 inflation rate correspond to £14.8 billion over the crisis year (the final step is linear in the rate; the non-uprating parameter changes in the two-vintage difference pull in both directions, the benefit cap, work allowances and childcare caps upward and the July 2022 National Insurance threshold downward, so the sign of the resulting bias is not determined), against the roughly

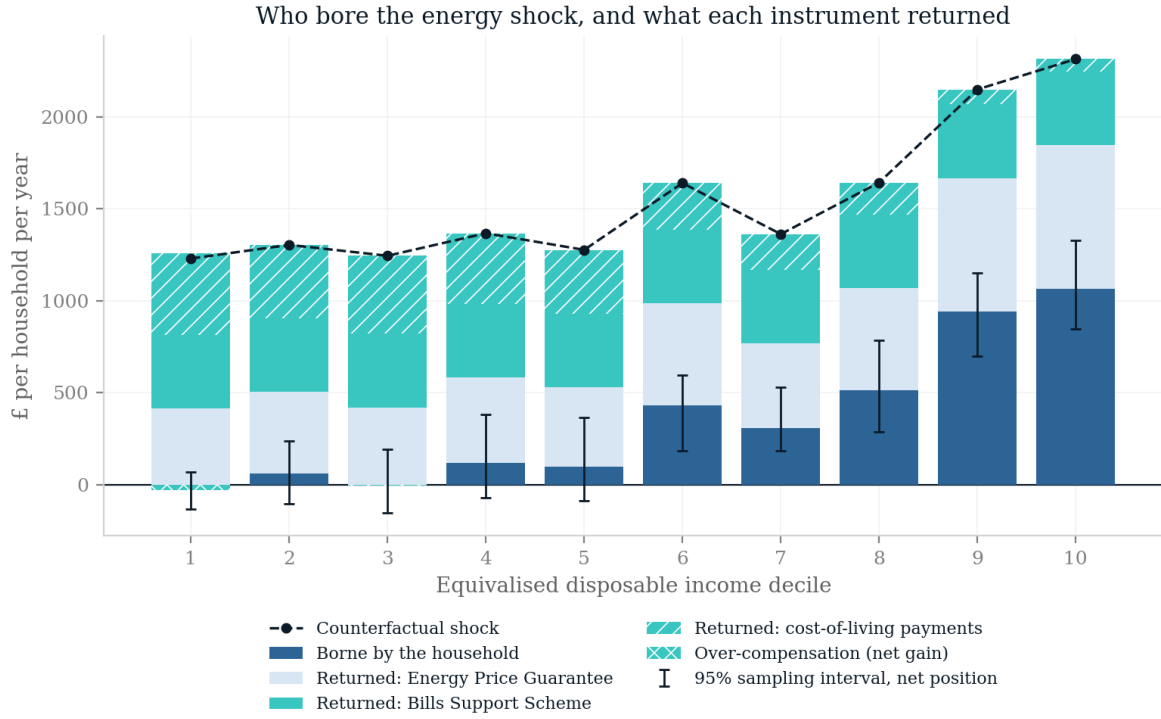
£12 billion aggregate real fall of OBR (2022b) on a different base and counterfactual, and below the £17.8 billion a flat multiplication of the 2023-rules base (£255 billion) would give, because tapers claw back part of any uprating. It is deliberately *not* expressed as a share of the energy shock: the numerator is the erosion of benefit income by all inflation, and dividing it by one commodity’s counterfactual would attribute to energy what food, services and rents also did. What the number establishes is narrower: the one automatic price-side channel, indexation, delivered its April 2022 step below the inflation of the year in which the shock landed.

5.4 The discretionary response by income decile

Against a counterfactual shock of £48.7 billion (128.6 per cent on the FY basis), the *energy-specific* instruments, the Guarantee (£16.5 billion) and the Bills Support Scheme (£12.6 billion), returned 59.5 per cent (66.3 per cent at the bottom decile, 51.1 at the top). Adding the cost-of-living payments (£8.7 billion) takes the full stack to £37.7 billion, or 77.4 per cent. Two attributions are reported: the cost-of-living payments were legislated against the general inflation episode, not energy specifically, so dividing them by the energy-only counterfactual over-attributes to this shock by the same logic on which the preceding subsection does not divide the indexation shortfall by the energy shock. On either attribution the rate falls with income: on the full stack, 102 per cent of the bottom decile’s counterfactual loss was returned against 54 per cent of the top decile’s, because flat payments are worth more relative to a small bill and the means-tested payments concentrate in the lower deciles. Two features of the profile deserve stating rather than smoothing. First, the bottom decile’s rate above 100 per cent (102 per cent, and 100 for the third decile) exists only on the full-stack attribution, the bootstrap below puts it within sampling error of 100 per cent, and it says only that the instruments returned about as much as the counterfactual energy cost, not that those households were made whole against inflation generally, which the preceding subsection shows they were not. Second, the gradient is not monotone: the sixth decile (74 per cent) sits below the fifth. Adding the Bills Support Scheme and the cost-of-living payments changes the pattern from the Guarantee’s constant rate to rates that fall with income. Figure 6 shows the anatomy in pounds: the counterfactual shock by decile, split into the part the household bore and what each instrument returned. A household bootstrap (999 resamples of the record base, following the practice urged for microsimulation results by Goedemé et al. (2013)) puts a 95 per cent sampling interval of 74.7–80.5 per cent around the 77.4 per cent aggregate rate. That interval bounds sampling noise, not specification error, and is narrower than the disclosed modelling approximations. The largest approximation is testable: recalibrating the EBSS to its £11.7 billion NAO outturn moves the full-stack rate from 77.4 to 75.6 per cent, so the headline survives it. The same bootstrap puts 94.8–111.8 per cent around the bottom decile’s 102: the bottom decile’s rate is therefore not statistically distinguishable from 100 per cent, and the net borne amount spans zero in each of the bottom five deciles (95 per cent intervals £-130–71, £-104–240, £-154–193, £-70–384 and £-86–367 a year for deciles one to five). These are sampling intervals over the survey base only: the resampling does not re-calibrate the grossing weights or reflect the survey’s design, and imputation uncertainty from the consumption donor models is not captured (Section 5.6).

5.5 The two-axis sensitivity grid

Table 4 varies the two conventions the headline rates depend on: the rebase factor (the paper’s computed 0.606, an intermediate 0.8, and 1.0, no rebase) and the discretionary stack, built up



Whiskers: household-bootstrap 95% sampling intervals on the net position; FY basis.

Figure 6: The anatomy of the energy episode in pounds a year by equivalised disposable income decile: the counterfactual shock (dashed line), the part borne by the household net of all modelled instruments, and the parts returned by the Energy Price Guarantee, the Bills Support Scheme and the cost-of-living payments. Whiskers are 95 per cent household-bootstrap sampling intervals on the net position; in the bottom five deciles it is indistinguishable from zero. Sampling uncertainty only; imputation uncertainty is not captured.

Table 4: Cushioning under two conventions: rebase factor $\times$ discretionary stack

Rebase	Stack	Shock (£bn)	Rate (%)	D1 (%)	D10 (%)
0.606 (paper)	None	48.7	0.0	0.0	0.0
	EPG only	48.7	33.8	33.8	33.8
	EPG + EBSS	48.7	59.5	66.3	51.1
	Full stack	48.7	77.4	102.5	54.0
0.800	None	64.3	0.0	0.0	0.0
	EPG only	64.3	33.8	33.8	33.8
	EPG + EBSS	64.3	53.3	58.4	46.9
	Full stack	64.3	66.8	85.8	49.1
1.000	None	80.4	0.0	0.0	0.0
	EPG only	80.4	33.8	33.8	33.8
	EPG + EBSS	80.4	49.4	53.5	44.3
	Full stack	80.4	60.2	75.4	46.1

Notes: Each cell is arithmetic on the same simulated households; the measured automatic response is zero in every cell (Section 5), so no re-simulation is required. “Rate” is the discretionary cushioning rate against the counterfactual shock on the financial-year basis; D1 and D10 are the bottom- and top-decile rates. The EPG row’s rate is constant across rebase and deciles by construction.

one instrument at a time. First, the Guarantee’s cushioning rate (33.8 per cent) is invariant to *both* axes, to the rebase because a price cap scales with the bill it truncates, and across deciles by the accounting identity already noted, so every pound of progressivity and every point of rebase-sensitivity in the headline comes from the flat-cash instruments. Second, across the grid the full stack cushions 77.4 per cent of the counterfactual shock at the paper’s rebase and 60.2 per cent with no rebase at all, and the progressive gradient survives every cell: without rebasing, the full modelled stack returns 75.4 per cent of the bottom decile’s loss against 46.1 per cent of the top decile’s. The direction of the results holds across the grid; the bottom decile’s rate above 100 per cent is the one result that is basis-dependent, present at the paper’s rebase (102.5 per cent) and absent without rebasing (75.4 per cent). One external anchor bears on the choice of cell: at a rebase of one the EPG absorption is £16.5 billion divided by the rebase factor 0.606, about £27 billion, close to the OBR’s £27 billion EPG costing, while the paper’s rebase gives £16.5 billion. The near-match is partly coincidental, the OBR priced the Guarantee against the higher caps then expected for 2023, over a longer window, on price-dependent volumes, but it cautions against reading the rebased cell as the truth and the unrebased one as a variant. The paper’s headline statements are therefore confined to what holds across the grid, and the basis-dependent rate above 100 per cent is reported as exactly that. The counterfactual *path* is also varied: holding the October 2022 cap through the financial year is a convention, not a forecast, so the grid is complemented by an *announced-path* variant using Ofgem’s November 2022 announcement of the January 2023 cap (£4,279), giving an FY mean of £2,942 and a counterfactual shock of £54.5 billion. On that path the Guarantee absorbs 40.7 per cent and the full stack 79.8 per cent: the paper’s held-cap convention gives the lower rate on the path axis and the energy-specific attribution the lower rate on the attribution axis; neither is a bound, because the rebase axis is a convention rather than an estimate.

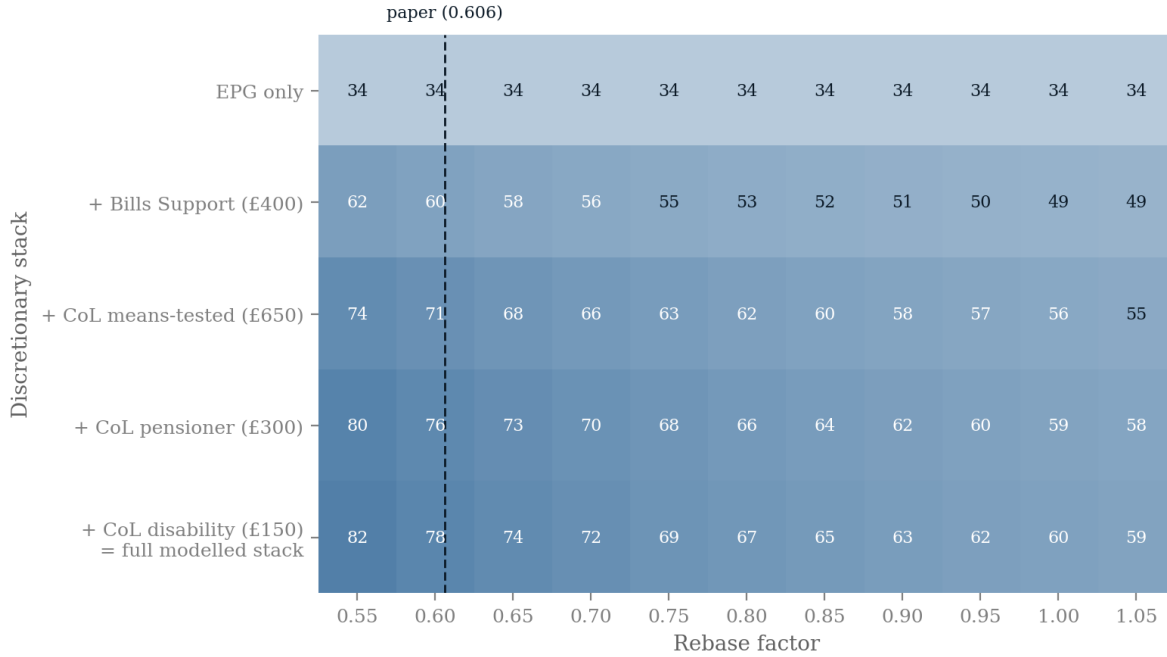


Figure 7: The two-axis sensitivity grid as a heatmap, with the rebase axis swept finely (0.55 to 1.05 in steps of 0.05) and the stack built one instrument at a time, with the cost-of-living payments decomposed into their three statutory components; the dashed rule marks the paper’s computed rebase, and Table 4 reports the three canonical columns with decile detail. Cell values are the discretionary cushioning rate against the counterfactual shock. The EPG row is invariant to both axes, a price cap scales with the bill it truncates and is constant across deciles by construction, so the progressivity and the rebase-sensitivity of the headline both come entirely from the flat-cash instruments.

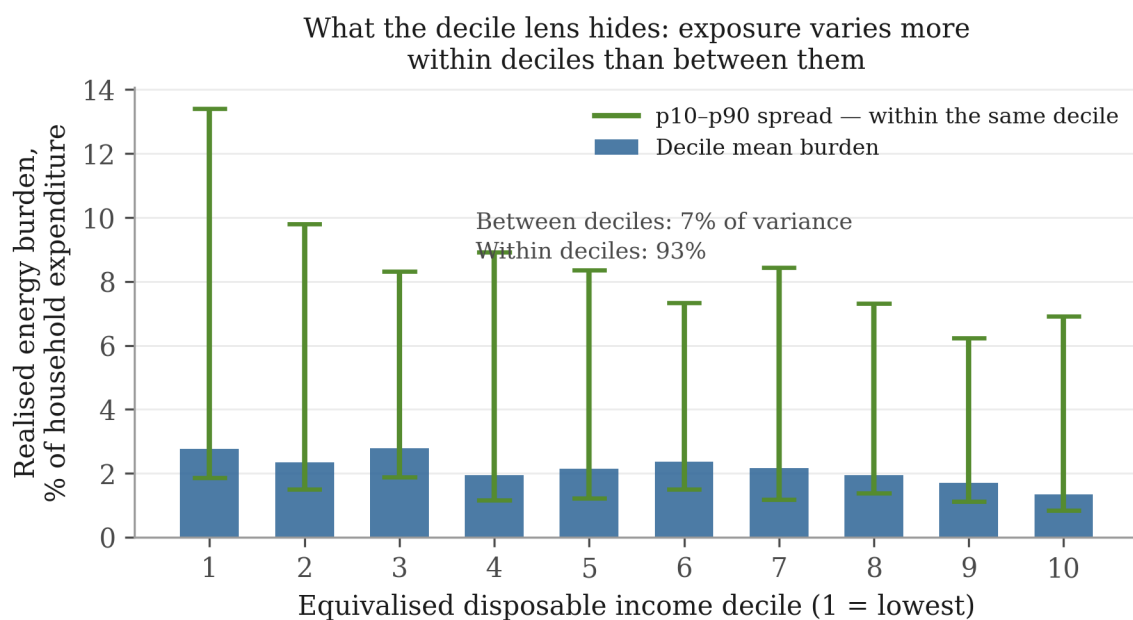
5.6 Exposure varies within deciles, not between them

Figure 9 plots the three shock paths of the design on one distribution: the gross counterfactual, the realised path, and the cash-outlay path (the gross counterfactual scaled by the observed demand response, the same convention as the first pass). The paths differ in level, not shape: policy and demand response scaled the shock down without changing who it fell on. The realised burden runs from 5.26 per cent of household expenditure in the bottom decile to 2.56 per cent in the top. The gradient is regressive and not monotone, the third decile (5.26 per cent) sits marginally above the first. On the unified base the levels sit close to their first-pass counterparts (5.26 against 5.93 per cent at the bottom decile), and energy exposure varies within income deciles as well as between them. Decomposing the variance of that burden puts 6.7 per cent of it between deciles and 93.3 per cent within them on the paper’s preferred basis (expenditure share, trimmed at the first and ninety-ninth percentiles; household-bootstrap 95 per cent interval 90.2–94.8 per cent for the within-share); untrimmed the between-share is 0.1 per cent and in logs 1.3 per cent. The trimmed expenditure-share figure is the paper’s estimate; the near-zero untrimmed between-share is dominated by a handful of extreme imputed shares in the tails and is reported as a data-quality diagnostic on the imputation, not as a robustness check. On every basis, though, the gradient the first pass measured is a small part of the dispersion in who was exposed. Within the bottom decile alone the burden runs from 3.52 per cent of spending at the tenth percentile to 25.20 per cent at the ninetieth.

Two limits bound what this establishes. The exposure is *imputed*, enhanced-FRS energy spending is a quantile-regression forest fitted to LCFS donors, so part of the within-decile dispersion is donor-model variation, and the decomposition describes the imputed exposure without establishing that it bounds the true dispersion, the bootstrap interval above is sampling uncertainty only, blind to the donor-model error, and a direct validation of the imputed within-decile spread against the LCFS donor sample, which observes it, is left to the extension; [Levell et al. \(2025\)](#) use panel data on household energy spending for this reason. And the estimand is the variance of one commodity’s expenditure share, not the variance of welfare changes from product-level trade exposure, which is what [Borusyak and Jaravel \(2024\)](#) decompose: the results point the same way, but they are not the same quantity.

5.7 A second episode through the second stage: the TCA food shock

The common cushioning estimand is computed for a second episode, and this one is a trade-*policy* act. The TCA food-price first stage (the window-average path of Section 4) is priced against each household’s imputed food consumption on the same 53,508 records, rebased so the weighted mean matches the first pass’s ONS FYE2022 food-spend base (factor 1.096; the same convention as the energy episode’s cap rebase, with shares on the same rebased spending base). The aggregate, £2.4 billion a year, therefore reproduces the first pass up to the grossing-weight excess; the new content of this stage is the distribution, not the level. The burden is regressive (0.35 per cent of bottom-decile spending against 0.21 at the top, against the first pass’s 0.35 and 0.20, matching at the bottom; the top-decile pound figure sits about a third above the first pass, a distributional divergence of the consumption imputation worth noting beside Section 5.6’s caveats) and larger in pounds at the top (£54 at the bottom against £128 at the top), and 95.9 per cent of the variance in exposure lies within deciles (trimmed basis; within the bottom decile the burden runs 0.22–2.14 per cent of spending at the tenth to ninetieth percentile). The cushioning decomposition is short: no modelled entitlement reads food consumption, so the cash-



Bars are each decile's mean realised 2022-23 energy burden as a share of total expenditure; whiskers span the 10th-90th percentile within the same decile. Equivalised deciles; enhanced FRS in PolicyEngine UK; variance shares on the trimmed expenditure-share basis. Consumption is imputed, so this is a bound.

Figure 8: What the decile lens hides. Bars are each decile's mean realised energy burden as a share of household expenditure; whiskers span the tenth to ninetieth percentile of that burden *within* the decile. On the trimmed expenditure-share basis the between-decile gradient accounts for 6.7 per cent of the variance in exposure and the within-decile spread for 93.3 per cent.

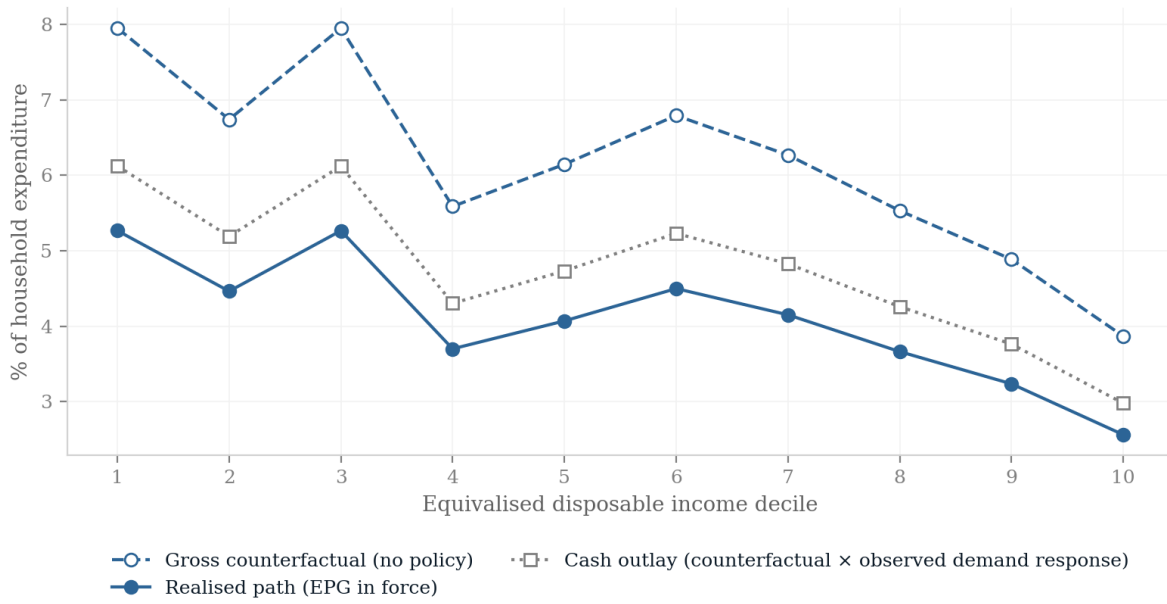


Figure 9: Three shock paths by equivalised disposable income decile, as a share of household expenditure, on the second stage’s financial-year basis: the gross counterfactual (no policy), the realised path (Energy Price Guarantee in force), and the cash-outlay path (the gross counterfactual scaled by the observed weather-adjusted demand response, the first-pass convention). Level shifts, common shape.

benefit response is zero by construction; indexation ran late, as above; and no episode-specific instrument exists, nothing was legislated against the TCA food-price effect. The contrast with the energy episode must be drawn on the episode-specific basis, by the same attribution logic as the discretionary subsection above: the general cost-of-living payments, which the episode-specific attribution excludes from the energy stack, also cushioned food-price inflation in 2022–23, so they cannot count toward energy and as zero for food. Episode-specific, the contrast is the paper’s central comparative fact: against the energy shock the state legislated instruments returning 59.5 per cent of the counterfactual loss; against the food shock, of the same sign and a similar spending-share gradient, it legislated none. The two episodes differed in size (the household-stage energy aggregate is 13 times the food aggregate) and in speed; the paper does not estimate why policymakers responded to one and not the other.

5.8 From incidence to demand and welfare

The statutory incidence measured above is an accounting object. This subsection aggregates it into two economic ones, keeping the paper’s conventions: every behavioural parameter is imported from the literature and declared, nothing is estimated, and the outputs are first-round policy-simulation quantities, no general-equilibrium feedback, no intertemporal response.

Demand. Each flow of the energy episode, the counterfactual burden and the three instruments, is weighted by a marginal propensity to consume assigned by equivalised-income decile, under three declared scenarios: a gradient falling from 0.8 in the bottom decile to 0.3 in the top, a deliberately steep stylisation of the cash-on-hand and liquidity gradients of [Jappelli and Pistaferri \(2014\)](#) and [Fagereng et al. \(2021\)](#), whose mapping into income deciles is flatter; a flat 0.5, which switches the heterogeneity channel off; and an asymmetric UK variant after [Bunn et al. \(2018\)](#), who find MPCs out of income *losses* (about 0.5) far larger than out of gains (about 0.14). Table 5 reports the results, with the flat scenario as the anchor and the gradient as the deliberately steep variant. First, under the gradient the counterfactual shock corresponds to a first-round consumption fall of £25.2 billion (£24.4 billion on the flat anchor); with the modelled stack the net first-round effect is £-4.4 billion. Second, under the gradient scenario only, first-round consumption per pound of modelled outlay is 0.63 for the cost-of-living payments, 0.55 for the Bills Support Scheme and 0.52 for the Guarantee; under the flat and asymmetric scenarios the three coincide, so the ordering is a property of the assumed gradient rather than an estimated one. Third, the asymmetric scenario applies [Bunn et al. \(2018\)](#)’s windfall gain-MPC of 0.14 to every instrument, although the Guarantee truncates a loss rather than paying a windfall; taken at face value the net first-round demand effect is £-19.1 billion.

Welfare. Weighting the counterfactual burden by [Atkinson \(1970\)](#) social-welfare weights on equivalised income (floored at £1,000 a year against near-zero survey incomes, a declared dial to which the $\gamma = 2$ premium is sensitive), the simplest member of the generalised welfare-weight framework of [Saez and Stantcheva \(2016\)](#), (proportional to $y^{-\gamma}$, normalised to mean one) gives a regressivity premium of 0.94 at $\gamma = 1$ and 1.23 at $\gamma = 2$. At $\gamma = 1$ the premium is below one: the shock is regressive in *shares* but larger in *pounds* at the top, so at log utility the pound-concentration at the top roughly offsets the share-concentration at the bottom, and only at higher inequality aversion does the welfare cost exceed that of an equal-pounds shock. Distributional statements about this episode therefore depend on the inequality aversion the reader brings, and the paper reports the mapping rather than choosing for them.

Table 5: First-round demand effects of the energy episode under declared MPC scenarios

Flow	First-round demand effect, £bn		
	MPC gradient	MPC flat	Fiscal flow
Counterfactual burden (no policy)	-25.2	-24.4	-48.7
Energy Price Guarantee	8.5	8.2	16.5
Bills Support Scheme	6.9	6.3	12.6
Cost-of-living payments	5.4	4.3	8.7
Net position (full modelled stack)	-4.4	-5.5	-11.0

Notes: MPC gradient: 0.8 (bottom decile) to 0.3 (top), linear; MPC flat: 0.5 everywhere. The asymmetric variant (Bunn et al., 2018: 0.5 out of losses, 0.14 out of gains) gives a net effect of £-19.1 billion (not shown by row). First-round, no general-equilibrium feedback; parameters imported, not estimated. Rates and sums are computed on unrounded arrays, so displayed rows may differ from column sums in the last digit.

6 Discussion and extensions

The household stage reports three things the first pass could only frame. With nominal incomes fixed, the modelled cash benefits do not respond to a price shock, because entitlements condition on nominal income; the model’s own Energy Price Guarantee variable does respond, and the paper counts it as discretionary. Indexation followed prices with a year’s lag, and the rulebook-vintage run puts the crisis-year difference against contemporaneous indexation at roughly 6.3 per cent of benefit income (£14.8 billion, a figure that also carries other parameter changes between the two rule years). The modelled discretionary instruments returned 49.4 to 63.8 per cent of the counterfactual energy shock across the rebase and price-path variants (60.2 to 79.8 per cent with the general cost-of-living payments attributed to it), at rates that fall with income. The second episode carried to household records sharpens this into the paper’s central comparative fact. The TCA food shock (price-led, a similar spending-share gradient, a policy act) had no instrument legislated against it; the two episodes differed in size and speed, and the paper does not estimate why the responses differed. And on the imputed exposure, most of the variance in the energy burden (93.3 per cent on the preferred basis) lies *within* income deciles; Levell et al. (2025) draw from the same pattern in household panel data the conclusion that an income-conditioned instrument targets energy exposure poorly.

Three descriptive findings hold across the paper’s variants. The negative shocks of the 2020s, two policy-made, one trade-transmitted, were regressive through prices, and the projected gains from the positive policy episodes are similar shares of spending across deciles and smaller in aggregate, because they arrive in categories (clothing, footwear) whose budget shares are flat across the distribution. The modelled responses ran through different mechanisms: the automatic tax–benefit system where the shock hit earnings (in the companion study’s scenario), discretionary instruments where it hit prices. The largest discretionary instrument, capping a price, cushioned proportionally by construction, and the full modelled stack on household records cushioned at rates that fall with income (Section 5). The welfare literature reads proportional energy support two ways: Levell et al. (2025) retain a positive price subsidy in their optimal package because income predicts energy exposure poorly, while Bonnet et al. (2025) find that transfers dominate subsidies for transport fuel, whose budget shares rise with income; the eight UK energy support schemes cost £44 billion against an initial £139 billion estimate, of which the Guarantee was £24.0 billion (NAO, 2024). A domestic incidence exercise cannot price the

externality that national subsidies impose through world energy demand (Auclert et al., 2023). Benefit indexation, the automatic price-side channel, followed the previous September’s inflation, so its April 2022 step fell below the inflation of the year in which the shock landed.

Past shocks carry ex-post estimates and recent agreements ex-ante projections. The sources reviewed for this paper include no published UK household-level ex-post validation of a trade agreement’s impact assessment; the Department for Business and Trade’s CPTPP assessment schedules an evaluation within five years of entry into force, and the Regulatory Policy Committee asked for the India assessment’s consumer analysis to be developed further (RPC, 2025). On the government’s own projections, the positive policy episodes are small and distributionally flat; on the published estimates, the negative episodes are large and regressive. A household-level ex-post analysis would put both on a comparable basis.

The limits are those declared throughout. First stages are imported, with their identification and their controversies; the incidence is first-order and static; the channels are not summed; the rulebook decomposition labels legislated change as discretion without asking what would have been legislated anyway. Two episodes have been carried to household records; the other three have not, and their first-pass numbers should be read accordingly. Poverty and inequality effects, the remaining episodes’ rulebook decompositions, and the consumption channel’s interaction with the earnings channel on common households all remain for the extension Section 6.1 specifies.

The framework can be reused: a future trade shock that fits its price or earnings channels enters as one more declared first stage for the same second stage, subject to the data and model limits stated above.

6.1 Extending the household stage

Section 5 runs the energy and TCA food episodes through PolicyEngine UK on the enhanced Family Resources Survey. Extending the full statutory stage to the remaining three episodes, and deepening the TCA run beyond its price vector, would add episode-specific parameters to the same second stage; this study has not implemented those runs, and four things would change when it is done.

First, the consumption channel moves from decile tables to household records: the enhanced FRS carries imputed household consumption across the twelve COICOP divisions plus the sub-items that matter most for trade shocks (petrol and diesel separately, in pounds and litres; electricity and gas separately; bus fares), imputed by quantile regression forest from the LCFS 2023–24 and calibrated to administrative energy and fuel totals. Each episode’s price vector therefore prices each household’s own basket, jointly with that household’s benefit entitlements, the covariance between exposure and cushioning that decile tables cannot see. Tariff-line and category-level first stages must be pre-aggregated to those roughly sixteen categories, and the imputation freezes budget-share *patterns* at the 2023–24 donor vintage across simulated years, a limitation the stage will state. Second, the earnings channels run through the companion study’s displacement machinery on the same households, so the two channels of a given episode land on a common population and their incidence can be compared household-by-household rather than distribution-by-distribution. Third, the rulebook axis would be scored directly: PolicyEngine’s parameter history carries dated values through the whole window, most with a legislative or Commons Library reference (the Covid-era Universal Credit uplift and its October 2021 removal, the March 2022 fuel-duty cut, the 2022 and 2023 upratings), its test suite exercises the 2020–2023 rule years, and a counterfactual-rulebook run applies an earlier year’s parameters to a later

survey vintage by copying the survey inputs to that year, as Section 4.3 does. Each episode is therefore scored under the rules in force at its date, under the preceding year’s rules (isolating uprating lag), and with the discretionary interventions switched off (isolating the automatic component), the three-way decomposition of Section 3 computed rather than approximated. Two verified limits shape the earlier years: no 2021-vintage enhanced-FRS data year exists (the earliest survey base is 2022–23), so 2021 rulebooks are applied to later households with incomes re-deflated; and the survey’s Universal Credit/legacy-benefit split follows reported receipt rather than the historical rollout, so 2021–22 rule-year results understate legacy-benefit interactions and are restricted to the tax and UC-parameter channels.

One methodological upgrade is specified here: the fixed-weight decile indices of the first pass can be replaced by non-homothetic distributional cost-of-living indices, which [Hochmuth et al. \(2025\)](#) construct from exactly the inputs this project already has (aggregate prices, expenditure shares and a single cross-sectional expenditure distribution), answering the standard objection that fixed baskets misprice large relative-price movements.

Two limits will survive the full stage. The consumption imputation carries its own error, inherited from the LCFS donor models, and imputation error correlated with benefit status would bias the exposure-cushioning covariance; the stage will report the imputation diagnostics alongside the results. And the rulebook decomposition attributes to “discretion” whatever the parameter history records as legislated change, which bundles genuine shock response with policy that would have happened anyway, a labelling choice, stated as such, not an identification claim.

Data and code availability

The pipeline, the pinned public inputs, the generated results and this manuscript are at <https://github.com/PolicyEngine/uk-trade-shock-study> (AGPL-3.0). The first pass uses only public inputs, each pinned by SHA256 and verified on every run. The second stage uses `enhanced_frs_2023_24.h5` from policyengine-uk-data release 1.55.12 (SHA256 584ae33d80ca0431254610a3f8254c97e987bb85eb03cc2cf0e15393e9101) derived from the Family Resources Survey under the UK Data Service licence and available through PolicyEngine’s data repository under the same licence; the model is policyengine-uk 2.95.0. The US-tariff episode’s three constants are taken from an unpublished companion manuscript whose pipeline is not part of this repository, and they should be read as scenario inputs until that manuscript and its code are public.

References

- Ahmadi, V. (2026). Who bears a trade shock? Wage cuts, job losses, and tax–benefit cushioning in the UK. Unpublished manuscript.
- Alabrese, E., Edenhofer, J., Fetzner, T. and Wang, S. (2026). Measuring the regional economic cost of Brexit: Evidence as of 2026. CAGE Working Paper, University of Warwick.
- Amaglobeli, D., Black, S., Celasun, O. and Toscani, F. (2026). Europe’s second energy shock: Avoiding the mistakes of 2022. *Intereconomics*, 61(3), 189–190.
- Amores, A. F., Basso, H. S., Bischl, S., De Agostini, P., De Poli, S., Dicarolo, E., Flevotomou, M., Freier, M., Maier, S., García-Miralles, E., Pidkuyko, M., Ricci, M. and Riscado, S. (2025). Inflation, fiscal policy, and inequality: The impact of the post-pandemic price surge and fiscal measures on European households. *Review of Income and Wealth*, 71(1), e12713. Earlier: ECB Occasional Paper 330 (2023).
- Ari, A., Arregui, N., Black, S., Celasun, O., Iakova, D., Mineshima, A., Mylonas, V., Parry, I., Teodoru, I. and Zhunussova, K. (2022). Surging energy prices in Europe in the aftermath of the war: How to support the vulnerable and speed up the transition away from fossil fuels. IMF Working Paper WP/22/152.
- Artuç, E., Porto, G. and Rijkers, B. (2019). Trading off the income gains and the inequality costs of trade policy. *Journal of International Economics*, 120, 1–45.
- Artuç, E., Porto, G. and Rijkers, B. (2021). Household impacts of tariffs: Data and results from agricultural trade protection. *The World Bank Economic Review*, 35(3), 563–585.
- Atkinson, A. B. (1970). On the measurement of inequality. *Journal of Economic Theory*, 2(3), 244–263.
- Auclert, A., Monneray, H., Rognlie, M. and Straub, L. (2023). Managing an energy shock: Fiscal and monetary policy. NBER Working Paper 31543.
- Bakker, J. D., Datta, N., Davies, R. and De Lyon, J. (2023). Brexit and consumer food prices: 2023 update. CEP Brexit Analysis No. 18, Centre for Economic Performance, LSE. Earlier: CEP Discussion Paper 1888 (2022).
- Bakker, J. D., Datta, N., Davies, R. and De Lyon, J. (2026). Non-tariff barriers and consumer prices: Evidence from Brexit. Working paper, March 2026; *Journal of Urban Economics*, forthcoming.
- Bank of England (2025). Monetary Policy Report, May 2025 (Box C: trade policy), and February 2026. <https://www.bankofengland.co.uk/monetary-policy-report/2025/may-2025>.
- Bonnet, O., Fize, E., Loisel, T. and Wilner, L. (2025). Compensating against fuel price inflation: Price subsidies or transfers? *Journal of Environmental Economics and Management*, 129, 103079.
- Borusyak, K. and Jaravel, X. (2024). Are trade wars class wars? The importance of trade-induced horizontal inequality. *Journal of International Economics*, 150, article 103935.

- Breinlich, H., Dhingra, S., Sampson, T. and Van Reenen, J. (2016). Who bears the pain? How the costs of Brexit would be distributed across income groups. CEP Brexit Analysis No. 07, Centre for Economic Performance, LSE.
- Breinlich, H., Leromain, E., Novy, D. and Sampson, T. (2022). The Brexit vote, inflation and UK living standards. *International Economic Review*, 63(1), 63–93.
- Brewer, M. and Tasseva, I. V. (2021). Did the UK policy response to Covid-19 protect household incomes? *The Journal of Economic Inequality*, 19(3), 433–458.
- Broadbent, B. (2022). The inflationary consequences of real shocks. Speech at Imperial College London, 20 October 2022, Bank of England.
- The Budget Lab at Yale (2025). State of U.S. tariffs (rolling distributional analyses of the 2025 tariff schedules). <https://budgetlab.yale.edu>.
- Calì, M., Maliszewska, M., Olekseyuk, Z. and Osorio-Rodarte, I. (2019). Economic and distributional impacts of free trade agreements: The case of Indonesia. World Bank Policy Research Working Paper 9021.
- Colombo, G. (2010). Linking CGE and microsimulation models: A comparison of different approaches. *International Journal of Microsimulation*, 3(1), 72–91.
- Bunn, P., Le Roux, J., Reinold, K. and Surico, P. (2018). The consumption response to positive and negative income shocks. *Journal of Monetary Economics*, 96, 1–15.
- Cohen, D. and Follette, G. (2000). The automatic fiscal stabilizers: Quietly doing their thing. *FRBNY Economic Policy Review*, 6(1), 35–67.
- Cravino, J. and Levchenko, A. A. (2017). The distributional consequences of large devaluations. *American Economic Review*, 107(11), 3477–3509.
- Curci, N., Savegnago, M., Zevi, G. and Zizza, R. (2025). The redistributive effects of inflation: A microsimulation analysis for Italy. *International Journal of Microsimulation*, 18(3), 87–100.
- Chen, T., Levell, P. and O’Connell, M. (2024). Measuring cost of living inequality during an inflation surge. IFS Working Papers W25/21 and W26/16, Institute for Fiscal Studies.
- Clarke, S., Serwicka, I. and Winters, L. A. (2017). Will Brexit raise the cost of living? *National Institute Economic Review*, 242, R37–R50.
- Davenport, A. and Levell, P. (2021). Brexit and labour market inequalities: Potential spatial and occupational impacts. IFS Working Paper W21/42, Institute for Fiscal Studies.
- Cribb, J., Joyce, R., Karjalainen, H., Ray-Chaudhuri, S., Waters, T., Wernham, T. and Xu, X. (2023). The cost of living crisis: A pre-Budget briefing. IFS Report R245, Institute for Fiscal Studies.
- Department for Business and Trade (2023). Impact assessment of the UK’s accession to CPTPP. <https://www.gov.uk/government/publications/cptpp-impact-assessment>.

- Department for Business and Trade (2025). Impact assessment of the Free Trade Agreement between the United Kingdom and the Republic of India. <https://www.gov.uk/government/publications/uk-india-free-trade-agreement-impact-assessment>.
- Dhingra, S. and Page, J. (2023). Accounting for imported and domestically generated inflation: Supply chains, monetary policy, and the UK’s cost of living crisis. VoxEU/CEPR, May 2023.
- Dhingra, S., Fry, E., Hale, S. and Jia, J. (2026). Counting the costs: A quantitative assessment of the effects of Brexit on UK regions and sectors. *Economica*, forthcoming, DOI: <https://doi.org/10.1111/ecca.70065>; earlier as CEP Discussion Paper 2187.
- Dolls, M., Fuest, C. and Peichl, A. (2012). Automatic stabilizers and economic crisis: US vs. Europe. *Journal of Public Economics*, 96(3–4), 279–294.
- Eckerstorfer, P., Riegler, M. and Sindermann, F. (2025). Inflation and government response: Distributional impact on Austrian households. *International Journal of Microsimulation*, 18(3), 132–154.
- Fajgelbaum, P. D. and Khandelwal, A. K. (2016). Measuring the unequal gains from trade. *Quarterly Journal of Economics*, 131(3), 1113–1180.
- Deaton, A. (1989). Rice prices and income distribution in Thailand: A non-parametric analysis. *Economic Journal*, 99(395), 1–37.
- Fagereng, A., Holm, M. B. and Natvik, G. J. (2021). MPC heterogeneity and household balance sheets. *American Economic Journal: Macroeconomics*, 13(4), 1–54.
- Ferreira, C., Leiva, M. J., Nuño, G., Ortiz, Á., Rodrigo, T. and Vazquez, S. (2026). The heterogeneous impact of inflation on households’ balance sheets. *SERIEs*, 17(1), 31–53 (Banco de España Working Paper 2403).
- Fleck, J. and Pradhan, S. (2026). The distributional effects of a US–EU trade war. Working paper.
- Figari, F., Paulus, A. and Sutherland, H. (2015). Microsimulation and policy analysis. In *Handbook of Income Distribution*, vol. 2B, 2141–2221. Elsevier.
- Freeman, R., Manova, K., Prayer, T. and Sampson, T. (2022). UK trade in the wake of Brexit. CEP Discussion Paper No. 1847, Centre for Economic Performance, LSE.
- Freeman, R., Garofalo, M., Longoni, E., Manova, K., Mari, R., Prayer, T. and Sampson, T. (2024). Deep integration and trade: UK firms in the wake of Brexit. CEP Discussion Paper 2066.
- Goedemé, T., Van den Bosch, K., Salanauskaite, L. and Verbist, G. (2013). Testing the statistical significance of microsimulation results: A plea. *International Journal of Microsimulation*, 6(3), 50–77.
- Hochmuth, P., Pettersson, M. and Jessen Weissert, C. (2025). A distributional cost-of-living index from aggregate data. *Review of Income and Wealth*, 71(4). DOI: <https://doi.org/10.1111/roiw.70034>.

- Jaravel, X. (2021). Inflation inequality: Measurement, causes, and policy implications. *Annual Review of Economics*, 13, 599–629.
- Jappelli, T. and Pistaferri, L. (2014). Fiscal policy and MPC heterogeneity. *American Economic Journal: Macroeconomics*, 6(4), 107–136.
- Judge, L. and Murphy, L. (2023). Rates of change: The impact of a below-inflation uprating on working-age benefits. Resolution Foundation, October 2023.
- Leventi, C., Mazzon, A. and Orlandi, F. (2024). Indexing wages to inflation in the EU: Fiscal drag and benefit erosion effects. EUROMOD Working Paper EM2/24.
- Levell, P., O’Connell, M. and Smith, K. (2025). The welfare effects of price shocks and household relief packages: Evidence from an energy crisis. IFS Working Paper W25/03 (updated W25/55), Institute for Fiscal Studies.
- Luu, N., et al. (2020). Mapping trade to household budget survey: A conversion framework for assessing the distributional impact of trade policies. OECD Trade Policy Paper No. 244.
- Navicke, J., Rastrigina, O. and Sutherland, H. (2014). Nowcasting indicators of poverty risk in the European Union: A microsimulation approach. *Social Indicators Research*, 119(1), 101–119.
- National Audit Office (2024). Energy bills support: An update. Session 2023–24, November 2024.
- Office for Budget Responsibility (2023). An international comparison of the cost of energy support packages. Box in *Economic and Fiscal Outlook*, March 2023. <https://obr.uk/box/an-international-comparison-of-the-cost-of-energy-support-packages/>.
- Office for Budget Responsibility (2022b). Benefit uprating during and after recessions. Box in *Welfare Trends Report*, May 2022.
- Office for Budget Responsibility (2025). Alternative trade policy scenarios. Box in *Economic and Fiscal Outlook*, March 2025.
- Office for Budget Responsibility (2026). Brexit analysis (standing forecast assumptions). <https://obr.uk/forecasts-in-depth/the-economy-forecast/brexit-analysis/>.
- Department for Energy Security and Net Zero (2024). Subnational electricity and gas consumption statistics: Regional and local authority, Great Britain, 2022. 25 January 2024. Weather-corrected domestic gas consumption fell 12.7 per cent in the gas year to mid-May 2023.
- Office of Gas and Electricity Markets (2021, 2022). Default tariff cap update from 1 October 2021 (6 August 2021) and from 1 October 2022 (26 August 2022), Figure 1: breakdown of default tariff cap components; with the accompanying *Default tariff cap level* model workbooks (v1.9, v1.13). <https://www.ofgem.gov.uk/energy-price-cap>.
- Office for National Statistics (2023). The purchasing power of GDP, UK: 2022. 8 February 2023.

- Pallotti, F., Paz-Pardo, G., Slacalek, J., Tristani, O. and Violante, G. L. (2023). Who bears the costs of inflation? Euro area households and the 2021–2022 shock. ECB Working Paper 2877.
- Offshore Energies UK (2023). UK energy import bills more than doubled to £117 billion in 2022. Press release, March 2023. <https://oeuk.org.uk/uk-energy-import-bills-more-than-doubled-to-117-billion-in-2022-and-could-hit-similar-highs-this-year-a-new-offshore-energies-uk-report-will-warn/>.
- Paulus, A., Sutherland, H. and Tasseva, I. (2020). Indexing out of poverty? Fiscal drag and benefit erosion in cross-national perspective. *Review of Income and Wealth*, 66(2), 311–333.
- Popova, D., Richiardi, M. and van de Ven, J. (2026). From pandemic to cost-of-living crisis: The distributional impact of UK tax and benefit policies, 2019–2023. CeMPA Working Paper 1/26, Centre for Microsimulation and Policy Analysis.
- Peichl, A. (2008). The benefits of linking CGE and microsimulation models: Evidence from a flat tax analysis. IZA Discussion Paper No. 3715.
- Saez, E. and Stantcheva, S. (2016). Generalized social marginal welfare weights for optimal tax theory. *American Economic Review*, 106(1), 24–45.
- Resolution Foundation (2022). Poorest households are set to see inflation nearly a third higher than the richest. Press release and analysis.
- Robilliard, A.-S. and Robinson, S. (2005). The social impact of a WTO agreement in Indonesia. World Bank Policy Research Working Paper 3747.
- Regulatory Policy Committee (2025). Opinion on the UK–India FTA impact assessment (RPC-DBT-25056).
- Sologon, D. M., O’Donoghue, C., Linden, J., Kyzyma, I. and Loughrey, J. (2025). Welfare and distributional impacts of inflation: A decomposition approach. *Review of Income and Wealth*. DOI: <https://doi.org/10.1111/roiw.70010>.
- Tax Policy Center (2025). Description of the Tax Policy Center’s tariff models. <https://taxpolicycenter.org/projects/resources/description-tax-policy-centers-tariff-models>.